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Analyzing Coopetition Dynamics in Open-Source Ecosystems: A Social Network and Repository Mining Approach

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Zusammenfassung

Open-Source-Softwareökosysteme sind zu wichtigen Orten zwischenbetrieblicher Interaktion geworden, insbesondere im Bereich der KI-Infrastruktur, in dem Unternehmen an gemeinsamen technischen Komponenten zusammenarbeiten und zugleich in verbundenen kommerziellen Märkten miteinander konkurrieren. Diese Arbeit untersucht zwischenbetriebliche Koopetition in einem unternehmensgeführten Open-Source-Ökosystem am empirischen Fall PyTorch. Zu diesem Zweck entwickelt die Studie einen repository-basierten Ansatz der sozialen Netzwerkanalyse, der auf Grundlage von Repository-Spurdaten ein firmenbezogenes Kollaborationsnetzwerk aus gemeinsamer Datei-Ko-Modifikation rekonstruiert und Kollaborationsmuster für das Gesamt-Repository sowie für Kern- und Peripheriebereiche vergleicht. Die Ergebnisse zeigen, dass die zwischenbetriebliche Zusammenarbeit in PyTorch konzentriert, hierarchisch und vom jeweiligen Repository-Bereich abhängig ist, wobei der Plattform-Sponsor die dominante strukturelle Position einnimmt und andere Unternehmen unterschiedliche Netzwerkrollen aufweisen. Die Analyse zeigt außerdem, dass direkte Wettbewerber des Sponsors nicht weniger kollaborativ sind als Nicht-Wettbewerber, was darauf hindeutet, dass nachgelagerte Rivalität mit anhaltender vorgelagerter Interdependenz rund um ein gemeinsames Framework koexistieren kann. Insgesamt zeigt die Arbeit, dass unternehmensgeführte Open-Source-Ökosysteme durch Sponsor-Asymmetrie, differenzierte Unternehmensrollen und fortbestehende Zusammenarbeit rund um geteilte Infrastruktur geprägt sind.

Summary

Open-source software ecosystems have become important sites of inter-firm interaction, particularly in AI infrastructure, where firms collaborate on shared technical components while competing in related commercial markets. This thesis examines inter-firm coopetition in a vendor-led open-source ecosystem, using PyTorch as the empirical case. To do so, it develops a repository-based social network analysis that reconstructs a firm-level collaboration network from repository trace data through shared-file co-modification and compares collaboration patterns across the overall repository as well as its core and peripheral scopes. The findings show that inter-firm collaboration in PyTorch is concentrated, hierarchical, and sensitive to repository scope, with the platform sponsor occupying the dominant structural position and other firms exhibiting distinct network roles. The analysis also shows that firms competing directly with the sponsor are not less collaborative than non-competitors, suggesting that downstream rivalry can coexist with sustained upstream interdependence around a shared framework. Overall, the thesis shows that vendor-led open-source ecosystems are structured by sponsor asymmetry, differentiated firm roles, and continued collaboration around shared infrastructure.

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1 Introduction

Open-source software ecosystems have become strategically important settings in which firms not only innovate independently, but also interact through shared technological infrastructures. In many such ecosystems, companies contribute to common code bases while pursuing distinct commercial interests in related markets. This makes open-source environments an important setting for the study of coopetition, understood as the simultaneous presence of cooperation and competition between firms [BN95, BK00, BK14, CSL+20]. The phenomenon is especially visible in contemporary AI infrastructure, where open-source deep learning frameworks such as PyTorch and TensorFlow function not only as technical tools, but also as shared infrastructures around which large technology firms interact strategically [ODH+24, TAL+25].

At the same time, open-source ecosystems are not organized in the same way. Some are governed in relatively distributed or foundation-led forms, whereas others are shaped more strongly by a dominant host firm. In such vendor-led or company-hosted ecosystems, participation may be formally open while influence over governance, development direction, and strategic coordination remains unevenly distributed [ODH+24]. These asymmetries make vendor-led OSS ecosystems especially relevant for understanding how collaboration and competition are organized together. They also raise the question of whether inter-firm coopetition in such settings is better understood as a relatively flat and community-like phenomenon or as a more layered and structurally differentiated one.

This thesis addresses that question through the empirical case of PyTorch. PyTorch is a widely used open-source deep learning framework with relevance across both research and commercial AI development. It originated within Meta, later moved into the PyTorch Foundation under the Linux Foundation, and is widely discussed alongside TensorFlow as one of the most important open-source AI frameworks [ODH+24, TAL+25]. PyTorch therefore provides a suitable setting for examining inter-firm coopetition in a historically vendor-led and still sponsor-asymmetric OSS ecosystem. It combines strategic relevance, sustained corporate participation, and a shared technological base on which multiple firms depend while differentiating themselves through complementary downstream products and services [ODH+24, TAL+25].

Despite the growing importance of such ecosystems, an important gap remains in the literature. Prior research offers several relevant but only partly connected strands: theoretical work on coopetition and paradox, studies of OSS governance and selective openness, and methodological work using repository mining and social network analysis to map collaboration structures [CSL+20, LR20, ALT+23]. What remains limited is an empirical framework that brings these strands together for the analysis of inter-firm coopetition in vendor-led OSS ecosystems. In particular, existing work has limited ability to connect structural network measures with strategic positioning, to extend contextual insight to large repository datasets, to distinguish more strategically meaningful forms of contribution from flatter measures of collaboration, and to account explicitly for the asymmetries of a dominant host firm [ODH+24, TAL+25]. This thesis addresses that gap.

The goal of this thesis is therefore to develop and apply a repository-based social network analysis approach for examining inter-firm coopetition in the PyTorch ecosystem. Rather than relying on surveys or self-reported relationships, the study reconstructs collaboration from repository trace data and interprets the resulting firm-level network through a coopetition lens. In doing so, it examines how observable development traces can be used to reconstruct a structural representation of inter-firm technical co-presence and how that structure relates to sponsor asymmetry, ecosystem roles, and competitive proximity. To guide the analysis, the thesis addresses three research questions:

RQ1: How can repository mining and social network analysis be used to model the collaborative dimension of inter-firm coopetition in open-source ecosystems?

RQ2: How do firms' structural positions within the reconstructed network differ, and how can these differences be interpreted in relation to distinct ecosystem roles?

RQ3: How do collaboration patterns differ between firms that are direct competitors and those that are not?

Methodologically, the thesis adopts a repository-based social network analysis design. The empirical analysis reconstructs a firm-level collaboration network from commit-file modification events extracted from the PyTorch repository. Collaboration is operationalized through shared-file co-modification, and the resulting network is examined across overall, core, and peripheral scope variants. To support theory-informed interpretation, the structural network measures are complemented by a separate annotation layer that captures ecosystem roles, firm types, and competitive proximity without altering the underlying topology of the graph. This design makes it possible to study inter-firm collaboration patterns at scale while remaining transparent about the distinction between computational reconstruction and later interpretive overlay.

Within these boundaries, the thesis makes three contributions. At the theoretical level, it refines the understanding of coopetition in vendor-led OSS ecosystems by showing how sponsor dominance, role differentiation, and collaboration among rivals around shared infrastructure can be interpreted together at the network level. At the methodological level, it develops a transparent workflow that links repository traces, firm attribution, structural network reconstruction, scope comparison, and theory-guided annotation. At the empirical level, it provides a detailed firm-level analysis of PyTorch as a strategically important AI infrastructure ecosystem. Together, these contributions support a more systematic empirical analysis of coopetition in open-source environments while remaining cautious about the inferential boundaries of the design.

The remainder of the thesis is structured as follows. Chapter 2 reviews the theoretical foundations of coopetition, OSS ecosystems, platform asymmetry, and repository-based network analysis, and derives the research gap addressed in this study. Chapter 3 presents the methodological design, including case selection, data collection, identity stabilization, firm attribution, network construction, scope differentiation, and annotation logic. Chapter 4 reports the empirical findings for the three research questions. Chapter 5 discusses the findings in relation to the theoretical framework, outlines the interpretation limits of the design, and summarizes the theoretical, methodological, and empirical contributions of the thesis. Finally, Chapter 6 concludes the study and outlines possible directions for future research.

2 Background and Problem Analysis

The chapter begins by presenting the fundamental concepts and the empirical context of the thesis (Section 2.1). It next elaborates on the strategic rationality of coopetition through game-theoretic and paradox perspectives (Section 2.2) and applies these concepts to open-source software and platform ecosystems (Section 2.3). Section 2.4 provides an overview of the application of social network analysis and repository mining to analyze such ecosystems. Lastly, Section 2.5 summarizes these strands into a research gap and inspires the empirical method of this thesis.

2.1 Key Concepts and Empirical Setting

This thesis explores how companies collaborate and compete simultaneously within a large open-source software ecosystem, based on the empirical example of the PyTorch ecosystem. This section provides a brief introduction to the key concepts and the empirical context of the thesis before turning to the theoretical foundation.

2.1.1 Coopetition

The core strategic concept in this thesis is coopetition. Brandenburger and Nalebuff (1995) argue that strategy is not only about competing within a fixed market, but also about changing the game by considering the broader set of interdependent actors involved, including customers, suppliers, substitutors, and complementors. Later work in strategic management and industrial marketing defines coopetition more explicitly as simultaneous cooperation and competition between firms, initially often in dyadic relationships and later also in broader multi-actor settings [BN95, BK00, BK14]. Recent work further emphasizes coopetition as a strategy pursued by firms with the intent to create value, while also involving paradoxical tensions that require management [CSL^{+}20].

In this thesis, coopetition is understood in this sense. Firms can jointly create value by contributing code, knowledge, and technical resources to shared open-source software projects while still pursuing distinct commercial interests in surrounding markets. This view is consistent with research on open-source software ecosystems showing that firms, including market rivals, may collaborate on shared technical assets while pursuing different commercial objectives [NCS$^+$18, ZZM$^+$20, ODH$^+$24]. These industrial open-source projects therefore provide a relevant setting for this thesis. The next subsection turns to open-source software ecosystems as the organisational context in which these coopetitive relationships unfold.

2.1.2 Open-Source Software Ecosystems

Many industrial open-source projects are better understood as ecosystems rather than as isolated repositories. An open-source software (OSS) ecosystem can be understood as a set of interdependent users, developers, organizations, artifacts, and infrastructure interacting around shared code bases [ZZM$^+$20]. Empirical research on large OSS ecosystems such as OpenStack shows that multiple companies and communities co-create shared technical resources while also pursuing their own strategic goals [TMH16, NCH$^+$17, ZZM$^+$20].

This thesis considers PyTorch to be an OSS ecosystem: a shared technical foundation that attracts contributions from multiple companies with partly overlapping and partly competing commercial interests [ODH$^+$24]. Within such ecosystems, some actors occupy more central coordinating positions than others [ODH$^+$24]. These asymmetries are taken up in the next subsection through research on platform ecosystems and orchestrators.

2.1.3 Platform Ecosystems and Orchestrators

According to platform research, a strong platform sponsor often directs the evolution of core technology while retaining control over important sources of appropriation [Wes03]. In open-source settings, governance may be vendor-neutral, as in foundation-hosted projects, or concentrated in a single host company [ODH$^+$24].

Previous research on OpenStack, enterprise-oriented OSS, and OSS community strategy shows that firms vary in their participation patterns, structural positions, and ability to build influence within such ecosystems [TRG15, ZZM$^+$20, LRD19]. One well-known example of such platform-like OSS ecosystems is provided by deep learning frameworks such as TensorFlow and PyTorch [ODH$^+$24]. Their role as strategic infrastructure for contemporary AI is outlined in the next subsection.

2.1.4 AI Frameworks as Strategic Infrastructure

TensorFlow and PyTorch are examples of deep learning frameworks that function as important shared infrastructure in contemporary AI development [ODH+24, TAL+25]. These frameworks provide shared tools for building and training machine learning models and are used across both research and commercial settings [ODH+24]. Recent research on company-hosted AI OSS projects shows that they are not only technical artefacts, but also sites of strategic interaction, with large technology companies collaboratively building shared tools while competing in surrounding markets [ODH+24, TAL+25].

The code base has both strategic and engineering importance because multiple firms rely on the same framework while pursuing competing AI-related products and services [ODH+24, TAL+25]. This makes PyTorch relevant not only as a software project, but also as a setting in which coopetition between companies becomes visible through patterns of code collaboration. The next subsection therefore provides a short overview of the PyTorch ecosystem and its governance.

2.1.5 Empirical Setting: The PyTorch Ecosystem (Short Overview)

Originally maintained by Facebook AI Research at Meta, PyTorch is an open-source deep learning framework that was donated to the Linux Foundation's PyTorch Foundation in September 2022 [ODH+24]. It is commonly discussed alongside TensorFlow as one of the major open-source deep learning frameworks, and the two projects have long been associated with rivalry over industry adoption [ODH+24].

PyTorch is an appropriate case for examining coopetition in a historically vendor-led AI OSS ecosystem because it originated within a major technology company, later moved into foundation-based governance, and has continued to attract broad use and contribution activity [ODH+24]. In this sense, PyTorch combines a company-hosted origin with later foundation-based governance, while still providing a useful setting for examining sponsor asymmetry, competitive relationships, and firm roles within a shared code base.

With these concepts and the empirical setting in place, the next section turns to the theoretical lenses used to analyze coopetition in PyTorch.

2.2 Theoretical Foundations: The Strategic Logic of Coopetition

Building on the basic notion of coopetition introduced in Section 2.1, this section presents two strategic lenses that structure the analysis: the value-net and game-changing view of

BRANDENBURGER AND NALEBUFF (1995), and the paradox perspective on managing simultaneous competition and cooperation. Business strategy has often been described using the language of warfare and rivalry, framing markets primarily in competitive terms. In the mid-1990s, Brandenburger and Nalebuff popularized the term co-opetition and argued for viewing business strategy through both cooperative and competitive lenses [BN95]. Co-opetition encourages firms to consider both cooperative and competitive interactions and to look for ways of reshaping the game to improve their strategic position [BN95]. Long-term business success depends not only on competing within the existing game, but also on actively shaping that game [BN95].

2.2.1 The Structure of the Game: The Value Net

To visualize this business environment, BRANDENBURGER AND NALEBUFF (1995) introduced the Value Net, a schematic framework that places the firm among four key categories of actors: customers, suppliers, substitutors, and complementors. Figure 2-1 illustrates this Value Net and the vertical and horizontal interdependencies between these actors. This framework reveals two fundamental symmetries in the game of business: a vertical symmetry between customers and suppliers, and a horizontal symmetry between substitutors and complementors [BN95].

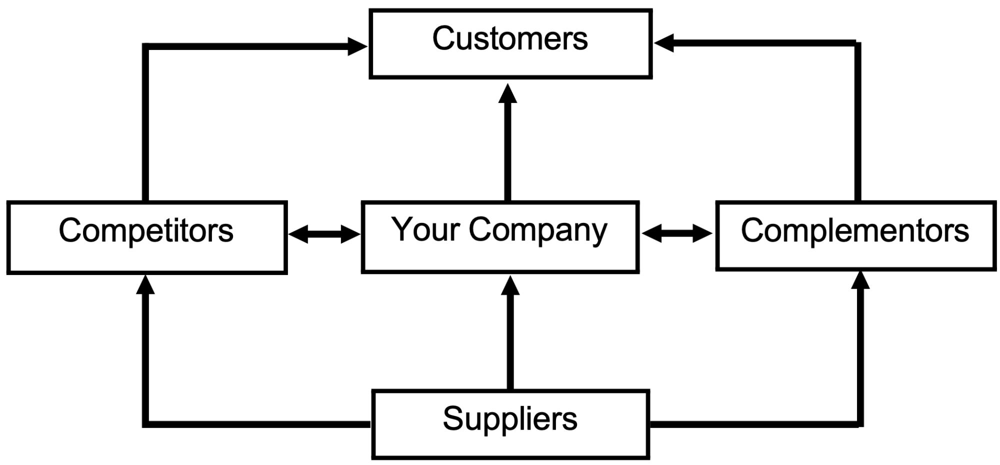


Figure 2-1: The Value Net framework illustrating the vertical and horizontal interdependencies that shift strategic focus from competition to co-opetition [BN95].

1. Suppliers provide resources that flow to the focal company and shape its position in the broader game [BN95].

2. Substitutors are players whose products or services make the focal company's offering less valuable [BN95].

3. Complementors provide products or services that add value to whatever the focal company offers (e.g., computer hardware and software) [BN95].

A central objective in this framework is to increase the firm's added value. Every value net has a total commercial value, and if a new participant joins, the amount by which the value of the whole net increases is that participant's added value [BN95]. Firms with higher added value are typically in a stronger strategic position within the value net [BN95]. While the Value Net describes the structure of the game, BRANDENBURGER AND NALEBUFF (1995) also show how firms can actively change the game.

They summarize such moves within the PARTS framework (Players, Added values, Rules, Tactics, Scope), which characterizes how firms can change who plays the game, what each actor brings, how interaction is structured, and where the boundaries of the game lie [BN95]. For this thesis, the key implication is that firms can reshape both the structure of the game and the set of participating actors. For example, firms may open a platform to external complementors and developers or open-source internal technologies in ways that broaden participation and enable collaboration in a shared ecosystem, even among firms that also compete elsewhere [Wes03, TAL+25]. This provides a useful starting point for examining who participates in PyTorch and how those firms are connected.

2.2.2 The Evolution of the Construct: From Static Dyads to Multi-Level Networks

The understanding of coopetition has broadened over time from an early focus on dual relationships toward a more dynamic and multi-level construct [BK14, CSL+20]. Early studies mostly examined coopetition from a dyadic perspective, focusing primarily on horizontal relationships between direct rivals [CSL+20, BK00]. This perspective has increasingly been seen as insufficient for more dynamic and interconnected business environments [BK14].

In a comprehensive review, BENGTSSON AND KOCK (2014) argue that the horizontal distinction is limiting [BK14]. They propose a refined definition: coopetition is a "paradoxical relationship between two or more actors, regardless of whether they are in horizontal or vertical relationships, simultaneously involved in cooperative and competitive interactions" [BK14]. This shift is particularly useful for analyzing software ecosystems, where firms may simultaneously occupy vertical and horizontal roles [BK14].

Additionally, researchers now stress the value of a multi-level viewpoint, emphasizing that coopetition is not only a firm-level strategy but a phenomenon that spans several levels [BK14, CSL$^+$20]:

1. **Inter-organisational Level:** Firms engage in coopetition to access external resources, develop innovation, and strengthen their competitive position [BK14].
2. **Intra-organisational Level:** Individuals inside firms may experience tension when cooperation and competition must be managed simultaneously [BK14, BK00].
3. **Network Level:** Coopetition can extend beyond dyads and therefore requires analysis of broader network or ecosystem structures [BK14, CSL$^+$20].

2.2.3 Open-Coopetition: Strategy in the Age of Open Source

A relevant development in the literature is the idea of open-coopetition, which highlights how competing firms collaborate within open-source ecosystems. In this sense, open-coopetition refers to situations in which firms that also compete in the market collaborate within a shared open-source project [TRG15].

In such settings, the dynamics of value creation and value capture differ from those of more closed collaborative arrangements. Open-source development is characterized by greater visibility of participation and contribution activity, which creates a distinctive strategic environment for inter-firm interaction. TEIXEIRA ET AL. (2015) use the OpenStack ecosystem to show that competing firms can collaborate visibly within the same project while still pursuing their own commercial interests [TRG15]. This suggests that, in open-source ecosystems, firms must think strategically not only about innovation itself but also about how participation and contribution shape their position within the ecosystem.

2.2.4 The Process of Coopetition: Managing Paradox and Tension

Whereas the game-based view of coopetition clarifies the structure of interdependence, recent management scholarship suggests that the simultaneous demands of cooperation and competition are often better understood through the lens of paradox theory [SL11]. The foundational work of SMITH AND LEWIS (2011) distinguishes a dilemma from a paradox. A dilemma presents an either-or choice in which one alternative is selected over another, whereas a paradox consists of contradictory yet interrelated elements that coexist and persist over time [SL11]. In software ecosystems, firms often face simultaneous pressures to share selected knowledge and code while also protecting strategically sensitive assets [LMW+22, LR20].

The Dynamic Equilibrium Model

SMITH AND LEWIS (2011) argue that organizations do not simply eliminate such contradictions, but manage them through a dynamic equilibrium of ongoing acceptance and resolution [SL11]. In this model, sustainability depends on continuous movement between complementary responses to tension:

1. **Splitting (Differentiation):** The physical or temporal separation of tensions [SL11]. In OSS settings, this may involve keeping differentiating or strategically sensitive parts closed while contributing commodity or enabling parts to the wider ecosystem [LMW+22, LR20].
2. **Integration:** The identification of synergies through which opposing demands can be accommodated together [SL11]. In OSS settings, this may occur when firms use openness to help establish standards, build ecosystems, or strengthen complementary commercial opportunities [LR20].

A closely related tension in this literature is that firms benefit from openness because it enables external knowledge inflows, collaboration, and ecosystem growth, yet they must also protect internal advantages and strategically sensitive assets [TMH16, ODH+24, LMW+22, LR20]. In open-source ecosystems, this tension becomes especially visible because development transparency and comparatively weak intellectual property restrictions can facilitate the transfer of information and resources across alliances, while also increasing the risk of spillovers and opportunistic behaviour [TMH16]. This paradox is particularly acute in vendor-led ecosystems. Empirical studies of AI OSS projects show that host firms must maintain enough control to coordinate the project strategically while also allowing enough autonomy to encourage external participation. Excessive control can discourage contributors, while excessive delegation can weaken strategic coordination [ODH+24]. Related work on open-coopetition further suggests that development transparency and weaker intellectual property restrictions can facilitate the transfer of information and resources across alliances, while also increasing the risk of spillovers and opportunistic behavior [TMH16].

In this context, research highlights the role of gatekeepers as an organisational mechanism for managing coopetition in OSS projects [NCS+18]. Gatekeepers are actors at the boundary between the firm and the open-source community who manage code and information flows across organizational boundaries, thereby helping firms balance value creation and value capture [NCS+18, NCH+17]. One way firms manage this boundary is through contribution filtering, deciding what code, information, and issues can be shared with the wider project and what should remain internal [NCH+17, LMW+22]. This perspective is important for this thesis because it frames boundary management not only as a protective mechanism but also as a strategic instrument. Because these tensions persist over time, firms must continually adjust how they manage the boundary between openness and strategic protection [SL11, LMW+22].

2.2.5 The Economic Logic: Market Expansion vs. Intellectual Property

Recent studies in high-tech platform ecosystems suggest that strict intellectual property protection is not always the most valuable strategic option [TAL+25]. TEIXEIRA ET AL. (2025) describe a critical trade-off between growing the market potential of a technology through open-sourcing and safeguarding market share through intellectual property protection [TAL+25].

Theoretically, releasing a complex platform as open source may involve giving up intellectual property, but this can be offset by an increase in the size of the overall market [TAL+25]. By lowering entry barriers and expanding adoption, an open platform can increase demand for complementary goods and services such as cloud computing and specialized hardware [TAL+25, Tya25]. According to this logic, under certain conditions an open ecosystem strategy may be more valuable to the sponsor than keeping the platform fully closed and proprietary [TAL+25].

2.2.6 The Source of Advantage: Network-Enabled Capabilities

To understand competitive advantage in ecosystems, it is useful to start from the Resource-Based View (RBV). In RBV, firm performance is explained by the internal resources and capabilities a firm controls [Lav06]. LAVIE (2006) extends this logic to interconnected firms, arguing that network resources, which firms do not fully own or control but can access through alliance partners, can be an important source of competitive advantage in alliance networks [Lav06]. This distinction between internal resources and network resources is essential for understanding coopetition in ecosystems, where firms combine access to shared external resources with the protection of strategically important internal assets.

However, access to these resources depends on the firm's position in the network structure [ZB05]. Social network theory helps explain how firms can benefit from advantageous network positions. ZAHEER AND BELL (2005) highlight the strategic importance of structural holes, understood as gaps in a network that allow a focal firm to connect otherwise disconnected actors or groups [ZB05]. Firms that bridge such gaps can act as information brokers and gain access to diverse, non-redundant knowledge that may enhance innovation and performance [ZB05]. By bridging structural holes, firms may be better positioned to exploit their internal capabilities and improve performance in coopetitive environments [ZB05]. Figure 2-2 illustrates this idea in a schematic way: the host firm acts as a strategic broker that bridges a structural hole between a community of contributors and commercial clients.

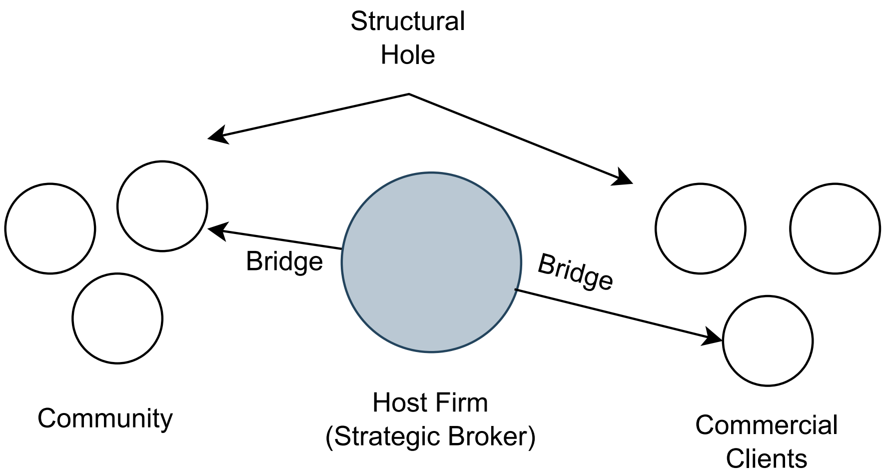


Figure 2-2: A host firm acts as a strategic broker by bridging a structural hole between disconnected groups. Conceptual illustration based on [ZB05].

Ideally, these internal and external perspectives should be integrated. Although the RBV highlights the significance of internal resources and social network theory emphasizes the importance of external ties, neither perspective alone fully explains competitive advantage in complex ecosystems [Lav06, ZB05]. ZAHEER AND BELL (2005) introduce the notion of network-enabled capabilities to bridge this gap. According to them, superior performance reflects a synergistic relationship between internal capabilities and network structure rather than dependence on either factor alone [ZB05].

A favorable network position is therefore important but not sufficient on its own [ZB05]. To realize this value, a firm also needs the absorptive capacity to process and apply the information accessed through its network [ZB05]. Their empirical evidence shows that innovative firms that also bridge structural holes perform better than firms that rely on internal capabilities or network position alone [ZB05].

This logic is particularly relevant for vendor-led ecosystems, where a host firm may occupy a structurally privileged position within the project and shape how external participation is organized [ODH+24]. In such settings, the host firm may occupy a strategic brokerage position between different contributors and organizational interests within the ecosystem [ODH+24, ZB05]. Its competitive edge may lie in the combination of a privileged structural position and the internal capabilities needed to integrate fragmented external inputs into a coherent offering [ZB05].

Related empirical studies of corporate-led OSS ecosystems support this broader logic. Analyzing OpenStack, ZHANG ET AL. (2020) show that a company's position in the collaboration network is positively associated with its productivity [ZZM+20]. This suggests that firms may gain tangible advantages when they occupy strategically advantageous positions within collaboration networks [ZZM+20]. Viewed through the lens of ZAHEER AND BELL (2005), these results suggest how firms can turn a favorable network position, together with internal absorptive capacity, into a source of competitive advantage within OSS ecosystems [ZB05, ZZM+20].

2.3 The Strategic Challenge: Coopetition in Vendor-Led OSS Ecosystems

Over the past two decades, OSS has increasingly become an important platform for industrial collaboration rather than being understood only as a volunteer-driven community activity. Earlier case studies of Apache and Mozilla had already shown that major OSS projects involved complex forms of distributed participation beyond purely volunteer contribution [MFH02]. Subsequent research shows that firms increasingly initiate and shape OSS projects in line with broader product and platform strategies, including through enterprise-led OSS and the commoditization of non-differentiating software components [SKK+20, VLM09]. To balance cooperation with competitive advantage, firms use OSS contribution strategies as a deliberate strategic tool for deciding what to share, when, and where [LR20, LMW+22]. Many important OSS projects are not neutral, community-governed entities but are instead organized as vendor-led or company-hosted ecosystems, in which a single commercial company initiates, hosts, and retains disproportionate control over the project's development, governance, and strategic direction [ODH+24]. This governance structure intensifies the coopetition dynamics described earlier: firms that are rivals in downstream markets nonetheless collaborate upstream on the same codebase, share development resources, and co-create a shared digital infrastructure [NCH+17, TMH16]. A vendor-led OSS project thus becomes a concrete arena where competition for market share and collaboration on core technical assets unfold at the same time. For example, in the AI ecosystem, external firms also contribute to PyTorch, a project initiated and long hosted by Meta [ODH+24].

The strategic challenge arises from the inherent asymmetry of power in this model. The host company is an active and often dominant participant rather than an impartial arbiter. In their study of company-hosted AI OSS projects, OSBORNE ET AL. (2024) report that up to 80% of the code may come from the host company. This allows the host to exert significant influence over the project's architecture, governance, and strategic direction, thereby shaping the rules of engagement in ways that may favor its own interests [ODH+24].

This creates a delicate balancing act for participating firms, both allies and rivals. In this context, "upstream" refers to collaboration on the shared open-source codebase, while "downstream" refers to the commercial products and services built on top of that code. Participating firms must work together upstream to influence technical direction, ensure downstream compatibility, reduce development costs through shared maintenance, and build influence in important parts of the ecosystem [TMH16, LR20, LMW+22]. At the same time, they must manage the downstream risks associated with competition. They therefore carefully select what they share, frequently employing internal gatekeepers: employees at the boundary between firm and community who screen contributions, decide what is revealed or kept proprietary, and translate between internal priorities and external expectations [NCS+18]. In parallel, many firms adopt selective openness strategies, releasing some parts of a product as open source while keeping advanced features, integrations, or services proprietary [Wes03, VLM09]. Together, gatekeeping and selective openness allow firms to stay active in the open community while protecting critical revenue sources. The primary strategic challenge in vendor-led OSS ecosystems is the conflict between upstream cooperation and downstream competition. As a result, firms must navigate a situation in which some of their closest competitors are also their partners, while one of those actors may set many of the rules of participation [BK00, CSL+20, ODH+24].

2.3.1 Governance Asymmetries and Power Imbalances: The Host's Advantage

Single-vendor governance creates risks that distinguish these ecosystems from foundation-led projects such as those under Apache or the Linux Foundation, where decision-making is more formally distributed across multiple stakeholders [ODH+24]. In vendor-led projects, a host company retains disproportionate decision-making authority, which makes external contributors dependent on the host for key roadmap, governance, and licensing decisions [ODH+24]. A key manifestation of this power is the possibility of unilateral licensing changes. Recent research on AI ecosystems shows that host firms may alter licensing terms to protect commercial interests, which can force external firms either to fork the project or to reconsider continued participation [ODH+24]. Moreover, the host shapes how far external companies are involved in day-to-day development, ranging from tightly controlled participation to more delegated forms of maintenance and contribution [ODH+24]. This creates uncertainty for participating firms, which must continually assess whether their influence within the project is sufficient to reduce the risk of lock-in, dependence, or strategic divergence. Compared with more meritocratic OSS communities, influence in vendor-led ecosystems is more strongly constrained by the strategic priorities of the host firm [ODH+24].

In practice, these governance asymmetries are also visible in who actually writes the code [ODH+24]. Figure 2-3 summarises the provenance of commits to PyTorch, TensorFlow and Transformers between 2016 and 2022. In all three projects, developers employed by the host company consistently account for the majority of commits, while contributions from external firms, volunteers and unknown affiliations remain comparatively small [ODH+24].

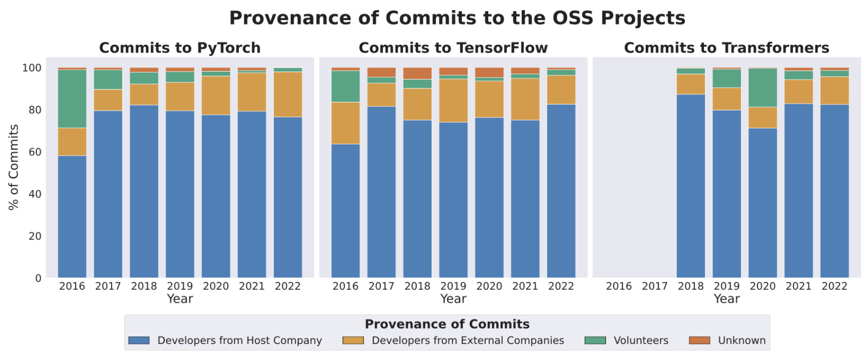


Figure 2-3: Provenance of commits to OSS projects per year [ODH+24]

2.3.2 Typology of Collaboration: How Rivals Interact

Coopetition in such ecosystems is not uniform, but takes several distinct forms of collaboration [ODH+24]. According to OSBORNE ET AL. (2024), there are three main ways through which host and external firms interact:

1. **Strategic Collaborations:** These are dyadic relationships between the host and selected external partners, such as hardware vendors or cloud providers. They are characterized by high levels of business alignment, intra-company communication, and joint roadmapping to ensure hardware-software compatibility [ODH+24].

2. **Contractual Collaborations:** In some cases, the host firm outsources specific development tasks to external consultants. These relationships are governed formally by contracts rather than open-source norms, even though the resulting code still becomes part of the shared commons [ODH+24].

3. **Non-Strategic Collaborations:** These include everyday maintenance work, bug fixes, and individual contributions by employees of competitor firms. Although less strategic, such interactions make up much of the daily coopetition within the ecosystem and blur the boundary between corporate and individual participation [ODH+24].

Defensive Coopetition: The Imperative to Join

A defensive form of coopetition arises when firms collaborate within a rival's ecosystem primarily to avoid strategic disadvantages rather than to pursue new opportunities. In dominant platform ecosystems, joining the shared project becomes a way to prevent loss of market share and to keep complementary products aligned with the emerging standard [TAL+25].

Participation in this form of coopetition is often driven less by voluntary alignment than by strategic necessity. TEIXEIRA ET AL. (2025) suggest that firms in a dominant ecosystem may be effectively pushed into open-coopetition in order to avoid losing market share [TAL+25]. This dynamic becomes visible when firms contribute to complex shared platforms in order to keep their complementary products aligned with the dominant ecosystem. For example, hardware vendors and cloud providers may contribute actively to a rival platform in order to maintain compatibility with their own complementary offerings [TAL+25]. Failure to participate may lead to exclusion from an ecosystem standard that evolves in ways that no longer fit a firm's proprietary infrastructure [TAL+25]. In this sense, open-coopetition can function as a defensive strategy within an ecosystem firms do not control, helping them protect downstream revenue streams [TAL+25].

2.3.3 The Economic Engine: Strategic Commoditization

Although governance structures establish the rules of the ecosystem, an important economic logic of vendor-led projects is strategic commoditization. In such ecosystems, open source can function not only as a collaboration mechanism but also as a strategic instrument for commoditizing non-differentiating parts of the market [VLM09].

The Commoditization Logic: The underlying idea is that software functionality often loses differentiating value as it matures and gradually becomes more commodity-like.

VAN DER LINDEN ET AL. (2009) emphasize that software technologies may lose much of their competitive value over time, and that firms should increasingly focus internal effort on differentiating parts while acquiring, sharing, or opening commodity functionality [VLM09]. In a vendor-led ecosystem, a host firm may accelerate this process by opening software that was previously kept internal, thereby speeding up the transition of some functionality toward commodity status. This can lower entry barriers and reduce the scope for rivals to sustain high margins on similar proprietary infrastructure solutions [VLM09].

Commoditizing Complements: This logic is related to recent economic modelling of complementary markets. TYAGI (2025) shows that firms with market power in one layer have strategic incentives to commoditize complementary layers that do not directly generate their primary revenues [Tya25]. For example, a firm may open a complementary layer of the stack in order to make it widely available, lower adoption barriers, and shift value capture toward proprietary services in another layer where it holds a stronger position [Tya25, TAL+25].

The Defensive Response: This dynamic can push smaller firms and rivals into a defensive position. Competing against a widely adopted, zero-priced open platform may be economically unattractive, so firms may join and help sustain the shared infrastructure rather than bear the cost and risk of maintaining a separate proprietary alternative [TAL+25, Wes03]. In this sense, participation in the ecosystem may become less a matter of strategic enthusiasm than of strategic necessity [TAL+25].

2.3.4 The Boundary Mechanism: Gatekeeping and Filtering

To manage these complex relationships, firms operationalize coopetition through internal procedures and decision processes. One essential mechanism is the use of gatekeepers, individuals who facilitate the flow of information and code between the firm's internal development teams and the external ecosystem [NCS+18, NCH+17]. These gatekeepers do not merely review code technically, but also play a strategic role in filtering what is shared so that differentiating intellectual property is not inadvertently disclosed [NCS+18, NCH+17].

LINÅKER ET AL. (2022) identify a fundamental practice known as business-driven filtering, in which firms sort software artifacts according to strategic value [LMW+22]. This aligns with the Contribution Acceptance Process (CAP) framework, which helps firms classify artifacts in terms of Business Impact and Control Complexity [LMW+22].

1. **Strategic Artifacts:** These have high business impact and high control complexity and are therefore usually kept proprietary or shared only under tight control [LMW+22].
2. **Commodity Artifacts:** These have lower differentiating value and are more readily contributed in order to reduce maintenance burden and support standardization [LMW+22].

Through systematic filtering mechanisms, firms can remain sufficiently open to influence the ecosystem and benefit from external innovation while remaining sufficiently closed to protect their competitive edge [LMW+22].

2.4 Current Tools for Analysis: A Review of Social Network Analysis in OSS Research

Social Network Analysis (SNA) has become an important methodological tool for quantitatively mapping and measuring relationships and flows between actors in a network [TFP+18]. These techniques have also been applied to OSS ecosystems in order to study complex collaboration patterns, including inter-firm interaction in projects such as OpenStack [TMH16]. Depending on the research design, the relevant actors may be organizations, individual developers, projects, or even technical artifacts. In OSS research, repository data such as commits and collaborative contribution traces can be used to infer relationships between these actors [ALT+23].

Early applications of these methods focused primarily on mapping the social structure of developer communities. Foundational studies of projects such as Apache and Mozilla revealed important patterns of governance, coordination, and contribution in large OSS communities [MFH02]. More recent work has expanded this focus by using SNA to map collaboration on a much larger scale. Temporal collaboration networks can now be built from millions of commits in order to trace changing influence and centrality across entire programming language ecosystems [ALT+23].

More recently, research has shifted beyond individual contributors toward the interaction of firms and other stakeholders. In the Apache Hadoop ecosystem, SNA has been used to study changes in stakeholder influence and collaboration patterns over time [LRR+16]. Inter-firm collaboration networks in large-scale projects such as OpenStack have also been explicitly visualized, providing an empirical basis for studying cooperation between competing firms [TMH16]. These developments show how SNA can move beyond developer-focused analysis and begin to capture the inter-organisational dynamics that characterize contemporary OSS.

2.4.1 SNA Fundamentals: Strategy as Metrics

SNA provides a quantitative way to describe collaboration by modeling actors as nodes and their relationships as ties [TFP+18]. Although these metrics originated in social network research, they can also be interpreted in ways that are useful for analyzing software ecosystems. SNA helps uncover how actors are connected, what structural roles they occupy, and how their position in the network can influence behaviour and outcomes over time [TFP+18]. Typical tasks include identifying influential actors, detecting communities, and tracing how information flows through the network structure [TFP+18].

To make these ideas more concrete, Figure 2-4 illustrates a typical collaboration network from an OSS ecosystem. Each node represents a developer or software artefact, and each edge denotes a collaborative tie such as co-editing files or co-authoring changes. Dense clusters correspond to communities or subsystems [TFP+18].

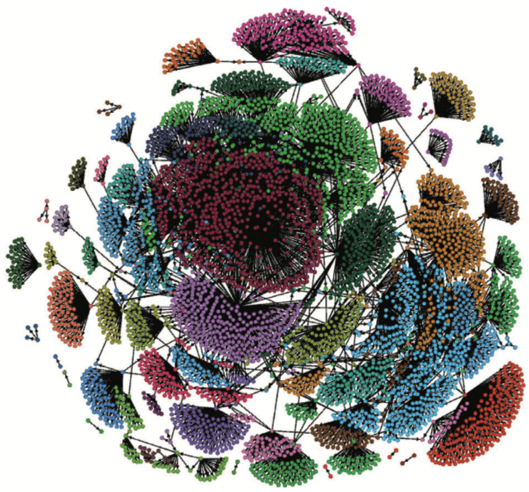


Figure 2-4: An example network visualization of an OSS ecosystem, where nodes represent developers or artifacts and edges denote collaborative ties [TFP+18].

Among the many available SNA measures, this thesis focuses on the metrics that are most useful for interpreting strategic positioning in OSS ecosystems, especially eigenvector centrality and betweenness centrality.

1. **Degree Centrality:** Degree centrality measures the number of direct ties of a node and captures its immediate connectedness within the network [TFP+18].

2. **Eigenvector Centrality:** Eigenvector centrality captures not only how many ties a node has, but also how well connected its neighbours are. It is therefore useful for identifying actors that are embedded in influential parts of the network and that may benefit from indirect access to important collaborators [TFP+18].

3. **Betweenness Centrality:** Betweenness centrality measures the extent to which a node lies on the shortest paths between other nodes. Actors with high betweenness can act as bridges or gatekeepers, influencing how information flows between otherwise weakly connected groups. This is especially relevant for identifying firms that bridge structural gaps and gain access to diverse, non-redundant information [TFP+18, ZB05].

2.4.2 Temporal Dynamics: Mapping Evolution Over Time

Early applications of SNA in OSS were primarily concerned with mapping the social structure of developer communities. Studies of Apache and Mozilla, for example, revealed important patterns in project governance, coordination, and contribution [MFH02].

Because collaboration structures evolve over time, static snapshots are often insufficient for capturing the dynamics of OSS ecosystems. Recent research by AGROSKIN ET AL. (2023) extends this approach through temporal network analysis, which slices repository history into specific time windows in order to examine how collaboration structures evolve [ALT+23]. Their work shows that it is possible to construct temporal collaboration networks from millions of commits in order to trace changing impact and centrality across entire programming language ecosystems [ALT+23]. Such methods can also help visualize how the relative influence and collaboration patterns of firms change over time, including possible strategic turning points [LRR+16].

More recent work has moved further toward explicit longitudinal analysis of inter-organisational networks. In their study of the TensorFlow ecosystem, TEIXEIRA ET AL. (2025) map the evolution of coopetition over a ten-year period by aggregating developer-level commit activity into firm-level nodes such as Google, Nvidia, and Intel [TAL+25]. This approach makes it possible to observe how ecosystem structures shift in response to external market developments, including moments when new competitors increase their participation in order to protect strategic interests [TAL+25].

2.4.3 The Aggregation Challenge: Bridging the Micro-Macro Divide

One key methodological issue in the study of vendor-led ecosystems is how to move from individual-level traces, such as commits, email identities, or pull requests, to firm-level conclusions about strategic behavior. Although OSS collaboration networks are often constructed from individual developer activity, the strategic unit of interest in industrial OSS is frequently the firm.

Aggregation Problem

There is no simple connection between these levels. Developers may contribute with personal email addresses, and organizational affiliations may change without a corresponding change in identity labels. SPINELLIS ET AL. (2020) emphasize that accurate attribution often requires non-trivial heuristics, including domain matching and cross-referencing with external datasets, in order to distinguish individual and organizational contributions [SKK$^+$20]. Without this kind of attribution work, firm-level strategic patterns can remain hidden behind developer-level activity [SKK$^+$20].

The Corporate Onion Model

When micro-level developer activity is aggregated to the organizational level, the resulting macro-structure can differ from the conventional onion model of OSS governance. In the traditional onion model, the core is often associated with highly active community contributors who gain influence through participation. However, NGUYEN-DUC ET AL. (2018) report that in firm-driven ecosystems, this structure is often reconfigured [NCS$^+$18]. Paid company actors such as gatekeepers and other firm-aligned contributors may dominate the core, while volunteers and competing firms are more often located in the periphery [NCS$^+$18]. Figure 2-5 (p. 20) illustrates this shift by contrasting the traditional volunteer-centric onion model with the firm-centric variant proposed for vendor-led ecosystems [NCS$^+$18].

Such a structural shift suggests that coreness in these ecosystems may reflect not only individual activity, but also the organizational influence of firms [NCS$^+$18]. This thesis follows earlier research that uses observable corporate affiliation and code ownership as the main lens for analyzing strategic behavior, despite the complexity of individual identities and motivations [SKK$^+$20, TMH16, ZZM$^+$20].

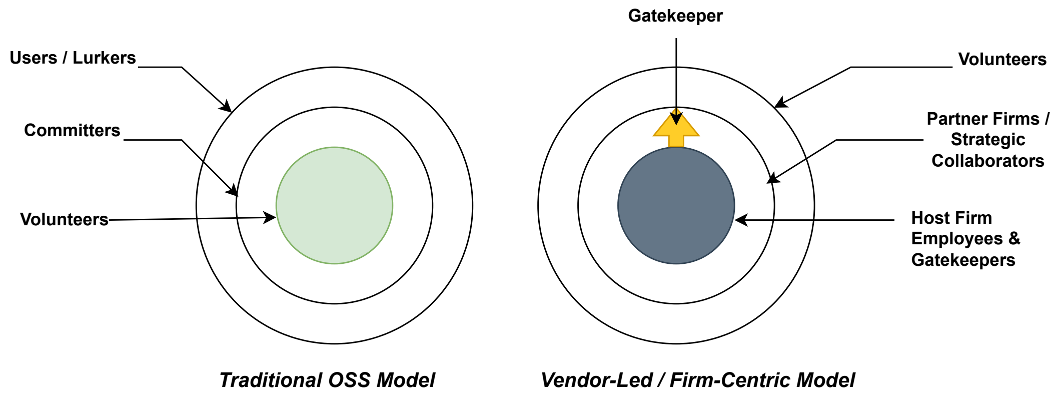


Figure 2-5: Comparison of the traditional volunteer-centric (left) and the firm-centric (right) Onion Models of governance [NCS+18].

Multilevel Analysis in Practice

To address these problems, recent research has gone beyond developer-oriented analysis to directly map inter-organisational dynamics. For instance, by aggregating individual developer contributions to their corporate affiliations, researchers were able to trace stakeholder influence in the Apache Hadoop ecosystem rather than only developer activity [LRR+16]. Similarly, OpenStack's inter-firm collaboration networks have been used to depict how companies cooperate within a shared ecosystem while competing elsewhere [TMH16]. These methods show how SNA can represent the multi-level reality of industrial OSS and move beyond the micro-level [LRR+16, TMH16, ZZM+20].

2.5 Research Gap: The Missing Link in Ecosystem Analysis

Although the methodological toolkit for studying open-source software ecosystems has advanced considerably, important gaps remain in the empirical analysis of inter-firm coopetition. Existing research provides valuable insights into collaboration structures, governance, and firm participation, but it remains fragmented and does not yet provide a sufficiently integrated empirical approach for capturing the strategic complexity of coopetition in vendor-led OSS ecosystems [CSL+20, TMH16, ODH+24]. A central issue is the gap between strategic theory and scalable empirical measurement.

2.5.1 The Divide: Strategy Theory vs. Empirical Software Engineering

A primary challenge is the disconnect between high-level theory and empirical measurement. The strategic management literature offers rich theoretical models of coopetition, including the value-net perspective and the management of paradoxical tensions [CSL+20]. However, much of this literature remains conceptual, review-based, or grounded in smaller-scale empirical designs rather than in large-scale behavioral data [CSL+20]. By contrast, the software engineering community has developed advanced techniques in Mining Software Repositories (MSR) and Social Network Analysis (SNA) for tracing interaction patterns in software projects [ALT+23, LRR+16, TMH16]. Yet many of these studies focus primarily on developer activity, community structure, or software-process outcomes rather than on strategic variables such as competitive positioning, governance asymmetry, or power dynamics between firms. As a result, while the topology of collaboration is increasingly well mapped, the strategic meaning of those ties remains difficult to assess systematically. Earlier studies that combine repository mining with contextual interpretation therefore point toward the value of approaches that connect quantitative rigor with richer strategic insight [LRR+16, TMH16].

2.5.2 The Trade-off: Contextual Depth vs. Analytical Scale

Beyond this disconnect between theory and measurement, a second challenge concerns the trade-off between analytical scale and contextual depth.

1. **Qualitative Limitations:** Many important insights into corporate participation in OSS come from qualitative case studies, interviews, and smaller-scale interpretive work. These studies have been especially useful for identifying mechanisms such as gatekeeping, selective revealing, and contribution filtering [NCH+17, LR20]. However, such approaches are difficult to scale across the large number of projects that make up contemporary OSS.
2. **Limitations of Scale:** At the other end of the spectrum, large-scale network studies can process millions of commits in order to construct temporal collaboration networks for entire ecosystems [ALT+23]. But such studies often stop at the level of developers, projects, or ecosystems rather than moving consistently to the firm level, where strategic interests are typically organized [LRR+16]. While newer datasets reveal the growing scale of enterprise-driven OSS, robust analytical models for interpreting such data strategically remain limited [SKK+20].

2.5.3 The Resolution Gap: The Problem of "Flat" Network Analysis

Even longitudinal research still faces a resolution problem in distinguishing strategically different forms of interaction. In particular, the TensorFlow study by TEIXEIRA ET AL. (2025) aggregates inter-firm interaction into weighted ties at the firm level [TAL+25]. This produces an important advance in scale, but under such a flat representation, strategically different forms of interaction may remain difficult to distinguish analytically. For example, high-value strategic collaboration and routine maintenance contributions can become mathematically comparable within the same edge weight.

TEIXEIRA ET AL. (2025) therefore rely on external qualitative accounts, such as interviews and public reporting, to interpret the strategic shifts visible in their networks [TAL+25]. This points to a gap between the strategic richness of coopetition theory and what current network metrics can quantify directly. Their interpretation suggests that some firms participate defensively to protect market position, whereas others participate more offensively to expand market opportunities [TAL+25]. Existing network methodologies, however, have limited ability to identify such strategic positions from repository data alone.

2.5.4 The Blind Spot: Asymmetry in Vendor-Led Projects

A further and more specific gap concerns vendor-led ecosystems. Existing work on OSS governance and coopetition has focused on projects hosted by vendor-neutral foundations such as the Apache Software Foundation or the Linux Foundation [ODH+24]. In such environments, governance is more formally distributed across multiple stakeholders.

By contrast, we still know comparatively less about coopetition in ecosystems that originate within, or remain strongly shaped by, a dominant company, such as Meta in the case of PyTorch or Google in TensorFlow [ODH+24]. Power, influence, and collaboration are structured differently in such projects because the host firm holds disproportionate authority over governance, licensing, and strategic direction [ODH+24]. Frameworks developed primarily for more neutral governance settings may therefore not transfer cleanly to these asymmetrical ecosystems. This leaves an important blind spot in our understanding of some of the most commercially significant AI OSS ecosystems [ODH+24].

2.5.5 Synthesis: The Need for a Unified Framework

Overall, the literature reviewed in this chapter shows that OSS ecosystems are increasingly important settings of strategic positioning, yet the research landscape remains fragmented. Existing work provides several important but still disconnected building blocks: theoretical models of coopetition and paradox (Section 2.2), governance mechanisms such as gatekeeping and selective openness (Section 2.3), and large-scale tools for network mapping and repository analysis (Section 2.4). What remains limited is an empirical framework that can bring these strands together for the analysis of inter-firm coopetition in vendor-led OSS ecosystems.

In particular, the current state of research still lacks an approach that can:

1. **Close the Theory-Method Gap:** connect quantitative network measures with richer indicators of strategic positioning and inter-firm tension.
2. **Scale Contextual Insight:** translate qualitative knowledge about selective revealing, contribution strategy, and firm behaviour to the scale of large repository datasets [LR20, ALT+23].
3. **Differentiate Contribution Types:** move beyond flat inter-firm ties by distinguishing contributions that are more likely to reflect strategic alignment from those that reflect routine maintenance [TAL+25, ODH+24].
4. **Capture Vendor-Led Asymmetry:** adapt network-based analysis to account for the power and influence of a dominant host firm in a company-hosted ecosystem [ODH+24].

These gaps point to the need for a scalable, data-oriented framework for analysing inter-firm coopetition in OSS ecosystems. This thesis addresses that need by applying the typological insights of Osborne et al. (2024) directly to the PyTorch ecosystem [ODH+24]. It also responds to Teixeira et al. (2025), who suggest that the strategic evolution of AI platform ecosystems cannot be fully understood without considering PyTorch alongside TensorFlow [TAL+25]. The goal of this thesis is therefore not to treat collaboration as a flat phenomenon, but to distinguish between strategically meaningful forms of alignment and more routine forms of maintenance within a vendor-led ecosystem. Empirically, the subsequent chapters develop a scalable method for operationalizing this distinction in the PyTorch project using repository and network data.

3 Methodology

3.1 Research Design and Empirical Strategy

This study employs a repository-based SNA design to examine inter-firm coopetition within the vendor-led open-source ecosystem of PyTorch. Rather than relying on surveys or self-reported ties, the analysis reconstructs collaboration from repository trace data. Prior OSS studies have shown that archival development data can be used to study collaboration structures among contributors and firms systematically [MFH02, ZZM+20].

The primary unit of observation is the commit-file modification event extracted from the PyTorch repository. Each event links a contributor identity, timestamp, and modified file path, thereby capturing granular technical engagement in the development process. The unit of analysis, however, is the firm rather than the individual developer. This firm-level perspective is consistent with prior OSS ecosystem research on inter-organizational collaboration and firm coopetition [LRR+16, NCS+18]. The identity stabilization and affiliation inference procedures are implemented as part of the present study's data-construction pipeline.

Within this design, collaboration is operationalized through co-change relations. Two firms are treated as connected when they modify the same file during the observation window, and tie strength is measured by the number of distinct files they co-modify. This study uses repository-based co-change as a structural proxy for technical co-presence in the ecosystem and as the basis for reconstructing the firm-firm network [ALT+23, TRG15].

The design further distinguishes between overall, core-only, and peripheral network variants. This distinction is introduced because not all repository activity reflects the same type of technical engagement. Separating the full repository from selected core paths and the remaining peripheral paths allows the analysis to compare whether inter-firm collaboration patterns differ across broader support layers and more foundational parts of the framework. The distinction is operationalized deterministically through file-path prefixes and is applied before network construction so that all three variants are built using the same edge logic. Figure 3-1 summarizes the empirical pipeline used in the study.

The empirical procedure begins with repository trace extraction within predefined temporal and branch boundaries. Contributor identities are then stabilized to reduce metadata

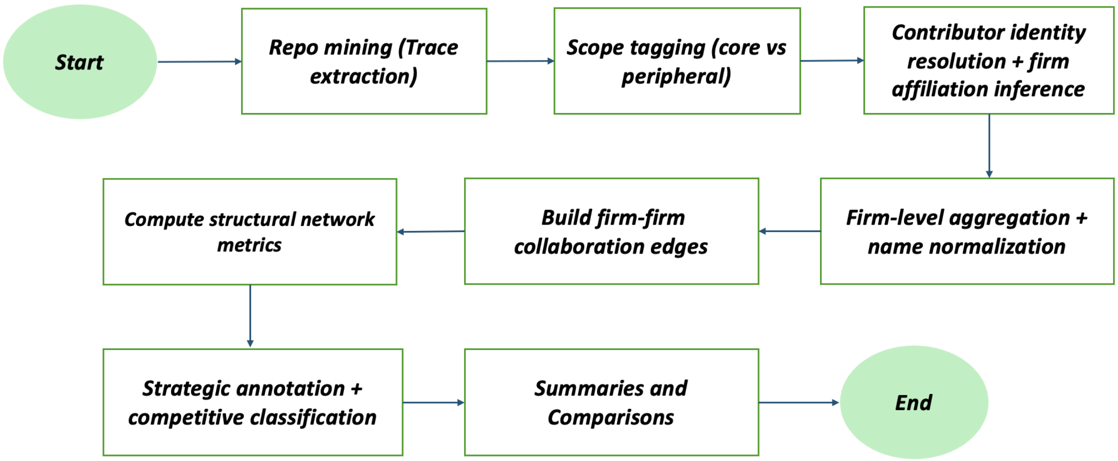


Figure 3-1: Empirical pipeline from repository traces to the firm-level coopetition network

fragmentation, after which firm affiliations are inferred using domain-based and, where necessary, profile-based signals. In the next stage, file-level activity is partitioned into core and peripheral domains using deterministic path rules. These prepared data are then used to construct weighted firm-firm networks based on shared file modification activity. Finally, structural network metrics and strategic annotation layers are added to support the empirical analysis of collaboration intensity, embeddedness, brokerage, and competitive positioning. The following sections explain each of these stages in detail.

3.2 Case Selection and Empirical Setting

The empirical context of this study is the PyTorch open-source ecosystem. PyTorch was selected after an exploratory screening of several candidate open-source ecosystems using a comparable preliminary contributor-affiliation procedure. Relative to the alternatives considered, PyTorch offered a strong empirical setting because it combined high observable firm participation with comparatively lower attribution ambiguity while also matching the substantive focus of this study on AI infrastructure.

The exploratory screening indicated that PyTorch contained a comparatively strong organizational signal for firm-level network reconstruction. In the preliminary results, firm-affiliated contributors accounted for a large share of both contributors and commits, while masked and uncertain contributor categories remained lower than in several other ecosystems considered during the screening. This made PyTorch a suitable case for reconstructing inter-firm collaboration patterns at the organizational level.

PyTorch also provides a strong substantive fit for the research question. It is a widely used open-source deep learning framework and can be treated as an important component of contemporary AI development infrastructure across research and commercial settings [AYH+24, TAL+25]. As in other AI platform ecosystems, a shared open-source framework can operate as a common technological base on which multiple firms depend while building differentiated complementary products and services around it [TAL+25]. This combination of shared upstream development and downstream differentiation makes PyTorch especially suitable for studying inter-firm coopetition.

In addition to its technological importance, PyTorch displays structural features associated with vendor-led, company-hosted open-source ecosystems [ODH+24]. Prior research indicates that large AI open-source frameworks rely heavily on sustained corporate participation and are often shaped by governance arrangements that structure how contributions and decision-making are coordinated [TAL+25, ODH+24]. Although contributions are made through open project workflows, prior research suggests that influence over development direction in company-hosted OSS projects may remain concentrated among a smaller set of corporate actors, especially the host firm and major contributors [ODH+24].

The ecosystem also reflects multi-firm dynamics that are characteristic of AI platform markets more broadly [TAL+25]. Competing firms collaborate in upstream development while simultaneously using the shared framework to differentiate themselves in downstream markets [TAL+25]. This combination of cooperation around a common technological base and competition in complementary commercial domains makes PyTorch a suitable empirical setting for studying coopetition.

3.3 Data Collection: Repository Mining and Trace Construction

3.3.1 Data Source, Trace Data, and Empirical Boundary

The empirical analysis is based on trace data collected from the PyTorch open-source repository (`pytorch/pytorch`) [PyT25]. In this study, trace data refers to the event records stored in the version-control history of the repository, especially commit metadata and file modification information. These records make it possible to reconstruct observable development activity from repository history rather than from surveys or self-reported relationships [MFH02, ZZM+20].

Collaboration is reconstructed from commit-file modification events. Each event links a commit to one modified file and can be associated with the author identity recorded in the commit metadata, the commit timestamp, and the file path. This representation provides the empirical basis for the later construction of contributor and firm-level relations.

Data extraction is limited to the repository's default branch, which is identified programmatically before the mining process begins, with a fallback to the `main` branch if the default branch cannot be resolved automatically. Restricting the extraction to the default branch ensures that the analysis reflects the repository's consolidated development history, that is, the branch in which accepted changes are integrated and maintained. This avoids mixing the main project trajectory with temporary, experimental, or unmerged work from side branches.

The observation window spans from 2014-06-01 to 2025-06-30, and these exact dates define the temporal boundary for all later aggregation and network construction procedures. The end date reflects the final cutoff used for data collection in this study. The start date was retained from the exploratory screening stage as a standardized lower temporal boundary across candidate ecosystems, which ensured consistency in the initial comparative setup before the final case was selected.

Merge commits are not explicitly removed from the dataset. As a result, the extracted trace reflects the complete development history of the selected branch within the defined time window. Retaining merge commits preserves the integrated branch history, including the points at which work from different branches is incorporated into the main project line. The advantage of this choice is that it captures the repository's full integration structure. A possible drawback is that merge commits may reflect integration activity rather than direct collaboration alone and can therefore increase the volume of observed change events.

3.3.2 Commit Traversal, Extraction Logic, and Reproducibility

Repository mining is carried out using PyDriller, a Python framework for mining software repositories that provides structured access to Git commit histories and related metadata. The extraction process traverses the locally cloned PyTorch repository on the selected branch and within the defined time period. The local repository state used for the present analysis was queried on 2025-11-24. For each commit, the procedure collects the author's name and email as recorded in the commit metadata, the commit timestamp, and the list of modified file paths.

This produces a detailed trace of development activity at file level by recording who made a change, when the change was made, and which file paths were affected. Trace-based reconstruction of this kind is an established approach in empirical software engineering and repository-mining research for studying collaboration and coordination patterns in version-control systems [MFH02, TRG15].

The extraction step produces two main relational datasets. The first is a commit metadata table containing commit-level information such as the commit hash and timestamp. The second is a commit-file table in which each modified file path is represented as a separate row linked to its originating commit. Keeping these datasets separate preserves file-level granularity and supports later steps such as contributor identity resolution, file-path scope classification, and firm-level co-change construction.

Several parts of this extraction stage follow fully deterministic logic, provided that the repository state and configuration settings remain fixed. These settings include the repository identifier, the selected branch, and the observation window. Commit inclusion is determined by the specified temporal boundaries, while modified file paths, author names, and author email addresses are retrieved directly from Git metadata without further transformation at this stage. Under the same repository conditions and configuration settings, rerunning the extraction procedure produces the same trace output. This deterministic setup provides a stable empirical foundation for the later construction of network measures and annotation layers.

3.3.3 Sources of Noise and Methodological Handling

Although the extraction procedure is deterministic, repository trace data are not free from structural noise. Version-control systems store author information as text fields rather than as verified identity records. As a result, one contributor may appear under multiple email addresses or slightly different name variants, while masked addresses such as GitHub-generated emails can obscure affiliation signals. At the same time, similar display names do not necessarily indicate that two commits belong to the same individual.

These characteristics can fragment contributor identities and complicate later attribution at contributor and firm level. In this study, such issues are not resolved during the mining stage itself. Instead, identity stabilization and affiliation inference are treated as separate methodological steps applied after trace extraction. This separation keeps the extraction stage focused on reconstructing observable repository activity, while subsequent stages convert that activity into more consistent contributor- and firm-level representations suitable for SNA.

3.4 Contributor Identity Resolution

3.4.1 Methodological Position and Canonicalization Rule

Building on the structural identity noise discussed in Section 3.3.3, the next step converts raw contributor traces into stable contributor entities. This stage is necessary because the analysis ultimately operates at the firm level, while the repository data are recorded at commit level and may contain fragmented contributor identities. Identity resolution therefore functions as an intermediate aggregation step between raw trace extraction and later firm-level attribution [ZZM+20, LRR+16, NCS+18]. In this study, the objective is not to solve the full entity-resolution problem, but to establish a stable and reproducible contributor representation that can support later organizational mapping.

To achieve this, the study adopts a deterministic consolidation rule based only on repository metadata. The identity resolution process is anchored on the exact author name recorded in the commit metadata. For each unique author name observed within the selected time window, all associated email addresses are collected, and a canonical email address is assigned by selecting the email that appears most frequently for that author name. All commit-file records linked to that exact author name are then updated to use the canonical email as the contributor identifier. When applied to the same repository data and observation period, this rule produces the same contributor mapping and therefore supports reproducible downstream aggregation.

The rationale for this choice is twofold. First, it reduces fragmentation caused by contributors using multiple email addresses over time. Second, it avoids more speculative forms of identity matching, such as merging across different author names on the basis of similarity or external assumptions. The method is therefore intentionally conservative: it resolves email aliases within an observed author-name identity, but does not attempt to infer whether different recorded names belong to the same person.

3.4.2 Design Trade-offs and Implications for Firm-Level Attribution

This design has important implications. Similar display names are not merged unless they appear under the exact same recorded author name in the repository metadata. As a result, the procedure reduces one source of fragmentation, namely multiple email addresses used by the same recorded name, but does not eliminate all possible identity ambiguity in the trace data. This is a deliberate trade-off in favor of transparency and reproducibility.

A second limitation is that the procedure does not attempt to model organizational shifts of contributors over time. Because the canonical email is selected as the most frequently used email within an author-name record, the stabilized contributor representation emphasizes the predominant identity signal observed in the dataset rather than temporal changes in affiliation. This choice is appropriate for the present study because the aim is to construct a stable firm-level collaboration network rather than to trace individual career mobility across organizations [LRR+16, NCS+18]. At the same time, it means that affiliation changes occurring within the observation window may be simplified in the final contributor representation.

At the company level, this approach improves consistency in the later aggregation of repository activity into firm participation patterns, because repeated contributions associated with one recorded contributor are less likely to be split across several email aliases. However, the conservative design may also understate cases in which one individual contributed under changing organizational affiliations during the observation period. The procedure should therefore be understood as a reproducible stabilization step that supports firm-level network construction, while not fully resolving all contributor-level ambiguity.

3.5 Firm Affiliation Inference and Organizational Attribution

After contributor identities are stabilized, contributions are mapped to organizational actors in order to construct a firm-level collaboration network. Because version-control repositories do not directly provide verified employer information, organizational affiliation must be inferred from observable digital signals rather than read directly from the repository data [SKK+20]. In this study, affiliation inference is implemented as a rule-based procedure that combines email-domain evidence as the primary signal with GitHub profile metadata as a secondary fallback signal. This distinction is important because the procedure contains both deterministic components and limited heuristic elements.

3.5.1 Primary Signal: Email-Domain Mapping

The primary affiliation signal is derived from the canonical contributor email address established in Section 3.4. For each contributor, the domain of the canonical email is extracted and matched against a curated corporate domain reference file. This file contains domain-company pairs and provides the basis for deterministic assignment whenever an observed domain is listed directly.

If a domain is part of the broader corporate-domain set but not explicitly linked to a named company in the mapping table, the procedure applies a standardized fallback rule that derives a provisional company label from the domain prefix. By contrast, generic personal email providers such as `gmail.com` or `outlook.com` are not treated as evidence of firm affiliation. Contributors using such domains are therefore not assigned to a firm on the basis of email alone. Given the same canonical identities and domain reference file, this stage produces reproducible domain-based assignments.

The curated domain reference file was compiled to support consistent organizational attribution across contributors. It was compiled manually by reviewing the domains observed in the repository data and linking them, where possible, to publicly identifiable corporate-domain associations. It serves as a structured lookup table rather than an ad hoc case-by-case assignment mechanism, which improves consistency in later aggregation.

3.5.2 Secondary Signal: GitHub Profile Metadata

To increase coverage where email-domain evidence is insufficient, the procedure also uses publicly available GitHub profile metadata. When a canonical contributor can be linked to a GitHub account and the profile contains a company field, that field is retrieved via the GitHub API and cleaned before further processing.

This profile-based signal does not replace the primary domain logic in every case. Instead, it mainly functions as a conservative fallback when the email signal is ambiguous, missing, personal, or masked. In practice, profile metadata can supply an organizational label in cases where email alone would otherwise yield a non-firm classification. Because the GitHub company field is self-declared and may be incomplete, outdated, or ambiguous, it is treated as a heuristic rather than fully deterministic source of evidence.

When neither domain-based nor profile-based information provides sufficiently credible firm evidence, contributors are retained in residual non-firm categories rather than being forcibly assigned to an organization. These residual categories include *Independent*, which refers to contributors using generic personal email domains without additional firm evidence; *Unaffiliated*, which refers to contributors whose domains are neither recognized corporate domains nor generic personal providers; and *GitHub (masked)*, which refers to GitHub-generated masked addresses such as `noreply.github.com`, where the email itself does not reveal organizational affiliation. Where direct firm evidence remains insufficient, the procedure may also assign intermediate uncertainty tags such as `likely_firm_affiliated` or `likely_independent`. These tags are generated through

Signal type	Illustrative example	Interpretation	Handling in this study
Corporate domain	name@huawei.com	Direct firm evidence	Assigned via domain map
Public provider	name@gmail.com	No firm evidence from email alone	GitHub fallback checked
GitHub masked email	id+name@users.noreply.github.com	Email does not reveal organization	GitHub fallback or residual
Academic domain	name@cornell.edu	Institutional, not commercial	Retained in output; filtered later

Figure 3-2: Illustrative examples of email and profile signals used in contributor affiliation inference

a simple activity-timing heuristic based on commit times recorded in the metadata and are intended to preserve ambiguous cases without forcing them into a firm label. Figure 3-2 provides illustrative examples of the main email and profile signal types encountered in the attribution procedure together with their interpretive handling in the study.

3.5.3 Contributor-Level Assignment, Normalization, and Network Inclusion

Affiliation is assigned at the contributor level. Once a canonical contributor is linked to an organizational label, all activities within the observation window are attributed to that contributor-level label. The procedure does not attempt to model employer changes over time. This static assignment supports consistent aggregation, although it may simplify cases in which a contributor changed organizational affiliation during the observation period.

Organizational labels are standardized before network construction to ensure consistent aggregation. Domain-based mappings provide the first layer of standardization, after which remaining label variants are normalized into a canonical firm identifier. This normalization step is important because equivalent organizational labels may otherwise appear under multiple written forms.

Not every contributor-level label becomes a firm node in the final network. The firm-level collaboration graph retains only contributors whose final classification is `firm_affiliated`. Residual non-firm categories and uncertain intermediate classifications, including `likely_firm_affiliated` and `likely_independent`, are retained in the intermediate attribution output for transparency but are excluded from firm-node construction. Academic institutions are also filtered later during firm-edge construction so that the final network remains focused on inter-firm relations.

Figure 3-3 summarizes the main decision logic used for contributor-level affiliation inference. For readability, repeated fallback steps are shown more than once in the diagram rather than being merged into a single crossing branch. The figure is intended as a simplified overview of the implemented logic rather than a line-by-line reproduction of every low-level decision step used in the code.

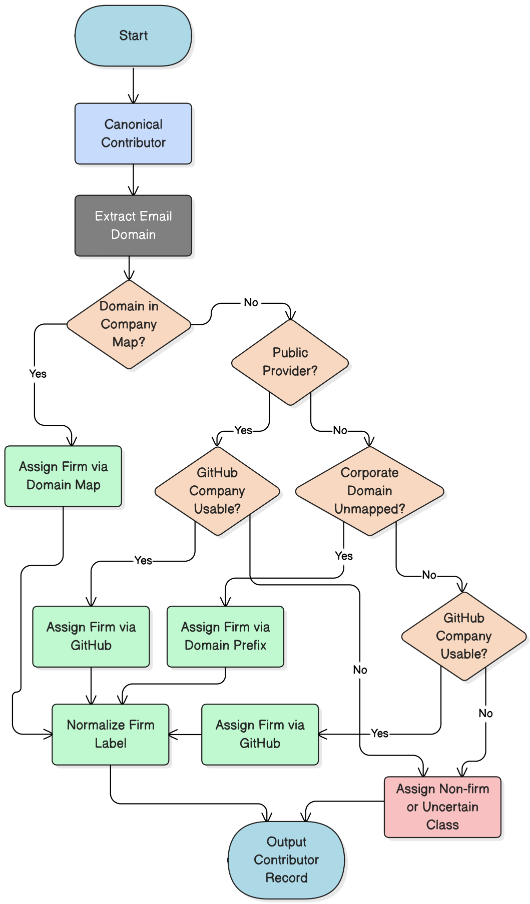


Figure 3-3: Summary of contributor affiliation inference using deterministic domain rules and conservative GitHub fallback

3.5.4 Residual Categories and Measurement Limits

Affiliation inference remains subject to measurement uncertainty. More generally, repository-based organizational attribution may be imperfect because observable project signals do not always map cleanly onto formal organizational roles [SKK+20]. In this study, possible sources of error include outdated or incomplete GitHub profile information, domains shared across subsidiaries or organizational units, and contributors acting outside formal employment roles. The rule-based design reduces these risks by separating deterministic domain assignments from heuristic fallback logic and by retaining residual categories where the evidence is insufficient. Even so, the procedure cannot fully eliminate the possibility of misclassification.

These limitations should be understood as measurement boundaries inherent to repository-based organizational inference rather than as failures of the overall design. In this study, the objective is not to establish legally verified employer identities for every contributor, which the underlying repository data do not directly provide. Instead, the aim is to construct a transparent, conservative, and reproducible firm-attribution layer that is sufficiently robust for downstream firm-level network analysis. This is supported by the fact that contributors without credible firm evidence are retained as residual or uncertain cases and are excluded from firm-node construction rather than being forcibly assigned to organizations.

3.6 Scope Operationalization: Core and Peripheral Contributions

3.6.1 Analytical Motivation

The firm-level collaboration network in this study is reconstructed from shared file modification activity. However, not all modified files reflect the same type of technical engagement. Large open-source repositories typically contain heterogeneous technical areas and contribution contexts rather than a single uniform type of development activity [TRG15, LRR+16]. Treating all file modifications as equivalent would therefore obscure potentially important differences in the nature of inter-firm collaboration.

To address this heterogeneity, the study operationalizes a scope-based distinction between core and peripheral contributions. The purpose of this study-specific distinction is not to claim a definitive ranking of technological importance for every file in the repository. Rather, it provides an analytical separation between contributions to selected foundational framework subsystems and contributions to surrounding support layers. This makes it possible to compare whether inter-firm collaboration patterns differ across different structural parts of the repository.

3.6.2 Deterministic File-Path Classification

Scope tagging is implemented as a deterministic rule applied to the file paths extracted during repository mining. Each modified path is checked against a predefined set of directory prefixes. A file is classified as *core* if its path begins with any of the following prefixes:

1. `torch/nn/`
2. `torch/_inductor/`
3. `torch/_dynamo/`
4. `torch/distributed/`
5. `torch/csrc/`
6. `aten/`
7. `c10/`

All remaining file paths are classified as *peripheral*.

The selection of these prefixes is informed by the repository's internal structure and on subsystem documentation that identifies these paths as central framework areas. For example, `c10/` is documented as part of the core backend architecture, while `torch/nn/` contains foundational neural network modules [PyT24c, PyT24a]. PyTorch's C++ API documentation likewise highlights components such as ATen and related backend-oriented building blocks, which supports treating directories such as `aten/` and other closely related implementation paths, including `torch/csrc/`, as part of the framework's foundational implementation layer [PyT24b]. The remaining selected prefixes, including `torch/_inductor/`, `torch/_dynamo/`, and `torch/distributed/`, are included because they correspond to major compiler and distributed-computing subsystems within the framework.

This classification rule is fully deterministic: the same path always receives the same scope tag. The procedure is therefore reproducible across the full dataset. At the same time, the chosen prefix set should be understood as an analytical operationalization for this study rather than as a claim to reproduce an official or exhaustive PyTorch architectural taxonomy.

3.6.3 Conceptual Interpretation

Under this operationalization, the selected core prefixes represent computational, compiler, distributed, and backend-oriented subsystems that form important parts of the framework's implementation base. By contrast, paths outside these directories typically include tests, documentation, examples, build configuration, auxiliary tooling, and other supporting layers of the repository.

The distinction between core and peripheral contributions therefore functions as an operational proxy for comparing collaboration around selected foundational subsystems with collaboration in surrounding support layers. Importantly, this distinction is generated through repository-structure rules rather than through case-by-case subjective coding. The value of the approach lies in providing a transparent and replicable partition of repository activity that can be used consistently in later network construction.

3.6.4 Methodological Boundaries and Limitations

The core-peripheral distinction should not be interpreted as a definitive measure of technological centrality. File-path prefixes are structural indicators derived from the organization of the repository, and they may not capture the precise architectural importance of every individual file. Some files outside the selected core prefixes may still be strategically important, while some files within the selected core directories may reflect routine maintenance rather than deep architectural work.

The extracted trace also records basic file-level event metadata such as change type. However, the present study does not use change type as a measure of contribution impact. More generally, the analysis does not operationalize impact through lines of code added or deleted, change size, or implementation progress. Instead, file modification activity is used as a structural indicator of participation and co-presence for the purpose of reconstructing inter-firm collaboration patterns.

Even so, the deterministic nature of the classification provides a useful methodological advantage. Because the partition is applied before network construction, the resulting overall, core-only, and peripheral-only networks are all built from a clearly defined and reproducible scope rule. This allows scope-specific comparison without changing the underlying tie-construction logic.

By separating scope operationalization from network measurement, the study preserves conceptual clarity between two distinct steps: first, partitioning repository activity into structural layers; and second, analyzing how inter-firm collaboration differs across those layers.

3.7 Firm-Firm Network Construction

After contributor affiliation inference and scope tagging, the analysis constructs a firm-level collaboration network in which firms are represented as nodes and ties are reconstructed from shared repository activity [TRG15, ZZM+20]. Commit-file records are aggregated to the firm level using the normalized contributor attributions described in Section 3.5. The resulting graph is modeled as an undirected, weighted network.

The unit of analysis in this stage is the firm pair. The purpose of the network is to capture whether and how strongly two firms are connected through shared modification of the same repository files. This design translates trace-level development activity into an inter-firm relational structure suitable for subsequent network analysis.

3.7.1 Edge Formation Rule: Shared-File Co-Change

An edge between two firms is created if both firms modified the same file at least once within the observation window. For each unique file path, the set of contributing firms is identified, and one tie is created for every unordered firm pair in that set. In other words, if multiple firms appear on the same file, all distinct firm pairs associated with that file are treated as connected.

Self-loops are excluded by construction. A self-loop would be a tie from a firm to itself, but the present network is intended to capture inter-firm relations only. The graph therefore contains ties only between distinct firms.

3.7.2 Edge Weight Definition

Collaboration strength is operationalized as the number of distinct files jointly modified by a firm pair. The edge weight between firms i and j is defined as:

$$w_{ij} = |\mathscr{F}_{ij}| \tag{3-1}$$

where $\mathscr{F}_{ij}$ denotes the set of distinct file paths modified by both firms i and j within the observation window.

Each file contributes at most one unit to w_{ij}, regardless of how many commits each firm made to that file. The resulting weight therefore reflects the breadth of shared artifact engagement rather than repeated activity on the same artifact. This is a deliberate design choice that reduces the influence of highly active single files on the overall edge weight.

The network does not use contribution size or technical impact as an edge-weighting rule. Although the trace includes basic file-level event metadata, the graph is not weighted by lines of code changed, commit volume on a file, or implementation progress. Instead, the weighting scheme is based solely on distinct shared files in order to capture structural co-presence between firms.

3.7.3 Firm Filtering and Organizational Scope

The firm-level network includes only contributors assigned the final tag *firm_affiliated*. Residual non-firm categories, including *Independent*, *Unaffiliated*, and *GitHub (masked)*, remain in intermediate attribution outputs for transparency but are excluded from firm-node construction. The same treatment is applied to uncertain intermediate tags *likely_firm_affiliated*, *likely_independent*.

To maintain focus on inter-firm coopetition, academic institutions are additionally filtered at the stage of firm-edge construction through name-based keyword matching (for example, *university*, *institute of technology*, and *hochschule*). This means that academic actors are not removed during contributor-level attribution itself, but are excluded when the final firm-firm graph is assembled.

3.7.4 Network Variants and Structural Properties

Three network variants are constructed using the same edge-formation and weighting logic: an overall network based on all eligible files, a core-only network, and a peripheral-only network based on the scope partition defined in Section 3.6. The rationale for this three-way design is analytical comparison. It allows the study to examine whether inter-firm collaboration patterns differ between the repository as a whole, selected foundational subsystems, and surrounding support layers, while holding the tie-construction logic constant.

Across all three variants, the resulting graph has the following structural properties:

1. **Undirected:** ties represent mutual participation in modifying a shared file rather than a directional sender-receiver relation.
2. **Weighted:** weights equal the number of distinct co-modified files.

3. **Simple weighted graph:** there is at most one edge per unordered firm pair, with intensity encoded in the edge weight.
4. **No self-loops:** ties exist only between distinct firms.

3.8 Network Metrics and Structural Operationalization

3.8.1 Analytical Objective

After constructing the firm-level collaboration networks, the analysis quantifies firms' structural positions using established SNA measures [TFP+18]. The purpose of these metrics is descriptive. They are used to characterize how firms are positioned within the collaboration structure, not to estimate causal effects. More specifically, the selected measures capture three complementary aspects of network position: collaboration intensity, embeddedness in the wider network, and brokerage potential between otherwise less directly connected firms.

All metrics are computed separately for the overall, core-only, and peripheral-only network variants. This allows the analysis to compare whether firms occupy similar or different structural positions across the repository as a whole, selected foundational subsystems, and surrounding support layers, while holding the underlying network-construction logic constant.

3.8.2 Collaboration Intensity: Weighted Degree

Collaboration intensity is measured using weighted degree [TFP+18]. In an undirected weighted network, weighted degree captures the total strength of a firm's ties by summing the weights of all edges connected to that firm [TFP+18]. In the present study, because edge weights represent the number of distinct files jointly modified by two firms, weighted degree indicates the breadth of shared technical engagement across all collaboration partners.

For firm i, weighted degree is defined as:

$$WD_i = \sum_{j \in N(i)} w_{ij} \tag{3-2}$$

where w_{ij} denotes the edge weight between firms i and j, and $N(i)$ is the set of firms connected to i.

For descriptive reference, the analysis also computes unweighted degree, that is, the number of distinct collaboration partners. However, weighted degree is the primary indicator used for collaboration intensity because it incorporates the breadth of shared artifact engagement rather than only the count of partners.

3.8.3 Embeddedness: Eigenvector Centrality

Embeddedness is operationalized using eigenvector centrality [TFP+18]. Whereas weighted degree captures the total strength of a firm's direct ties, eigenvector centrality captures whether a firm is connected to other structurally prominent firms [TFP+18]. A firm therefore receives a higher score not only when it has many or strong ties, but when those ties connect it to firms that are themselves well connected in the network.

For firm i, eigenvector centrality is defined as:

$$EC_i = \frac{1}{\lambda} \sum_j w_{ij} EC_j \tag{3-3}$$

where λ is the leading eigenvalue of the weighted adjacency matrix and w_{ij} denotes the edge weight between firms i and j.

The measure is computed on the weighted undirected network. If iterative numerical convergence is not achieved, a matrix-based eigenvalue solution is used as a fallback so that the metric can still be computed consistently.

3.8.4 Brokerage: Betweenness Centrality

Brokerage potential is measured using betweenness centrality [TFP+18, ZB05]. Betweenness centrality captures the extent to which a firm lies on shortest paths between other firms and can therefore occupy an intermediary or bridging position in the network [TFP+18, ZB05]. In substantive terms, it indicates whether a firm is structurally positioned between other actors that are less directly connected to one another.

For firm i, betweenness centrality is defined as:

$$BC_i = \sum_{\substack{s \neq i \\ t \neq i \\ s \neq t}} \frac{\sigma_{st}(i)}{\sigma_{st}} \tag{3-4}$$

where σ_{st} is the number of shortest paths between firms s and t, and $\sigma_{st}(i)$ denotes the number of those paths that pass through firm i.

In this study, shortest paths for betweenness centrality are computed on the unweighted graph. Accordingly, brokerage reflects topological positioning in the collaboration structure rather than the intensity of co-modification captured by edge weights. The measure is normalized, which supports comparison across network variants of different sizes.

3.8.5 Scope-Specific Computation and Interpretation

All three metrics are computed separately for the overall, core-only, and peripheral-only networks using identical procedures. This preserves comparability across scope partitions and ensures that any observed differences in structural position reflect the underlying repository partition rather than changes in metric definition.

These measures should be interpreted as structural descriptors derived from shared artifact engagement. They do not indicate direct contractual cooperation, formal alliances, or causal influence. Rather, they describe how firms are positioned within the observable inter-firm collaboration structure reconstructed from repository activity.

3.9 Strategic Annotation and Competitive Arena Classification

3.9.1 Analytical Position

The firm-level networks and structural metrics describe how firms are positioned within the PyTorch ecosystem, but they do not by themselves indicate how those firms relate to the broader strategic setting of a vendor-led open-source ecosystem. To support theory-informed interpretation within a coopetition perspective, this study adds a separate annotation layer that overlays externally assigned firm attributes onto the computed network metrics. The purpose of this layer is interpretive: it enables structural positions to be read in light of annotated ecosystem role and competitive proximity without changing the underlying network topology, node set, or edge weights.

3.9.2 Manual Annotation Framework and Role Taxonomy

The detailed annotation framework is applied only to a focal subset of firms. The weighted-degree ranking in the overall network is used to identify the most relevant firms for detailed strategic annotation. In practice, detailed coding is applied only where a sufficiently clear organizational identity and publicly observable market position can be established with confidence. This results in a final annotated subset of 54 firms.

For this focal subset, three detailed annotation variables are assigned manually:

1. **role_label**: fine-grained ecosystem-role classification within the PyTorch setting
2. **firm_type**: broader market-identity grouping outside the repository context
3. **direct_competitor**: binary indicator of downstream market overlap with the platform sponsor

At a broad conceptual level, the fine-grained **role_label** assignments can be interpreted in relation to a smaller set of ecosystem-role families used here to explain the annotation logic:

1. ecosystem orchestrator / platform sponsor
2. core hardware complementor
3. cloud platform complementor
4. foundation model lab
5. enterprise or domain-specific AI user
6. integrator or technical services firm
7. peripheral or specialized contributor

Here, the platform sponsor refers to the vendor-led host organization that stewards the PyTorch repository in the focal ecosystem setting [ODH+24]. The role taxonomy is a study-specific interpretive classification informed by publicly observable firm characteristics and by prior work on platform governance, sponsor relationships, and complementor roles in open-source AI ecosystems [TAL+25, ODH+24]. It is therefore theory-guided rather than automatically inferred from network position. In the context of this study, these role categories are treated as *strategic roles* because they capture firms' observable positions in relation to the platform sponsor, their mode of participation in the focal ecosystem, and their likely relevance for coopetitive interaction around the shared technological base. The term therefore does not refer to latent intentions or internally formulated corporate strategies. Rather, it refers to externally interpretable ecosystem positions that are strategically meaningful for understanding how firms simultaneously collaborate and compete in a vendor-led OSS setting.

For the actual manual coding of focal firms, these broader role families are differentiated into more fine-grained empirical labels where needed in order to preserve meaningful variation across firms' market positions and technical roles and are provided in Appendix A. These detailed labels should therefore be understood as nested subcategories rather than as unrelated standalone role classes.

The **firm_type** variable captures a firm's broader market identity outside the repository context, for example `gpu_vendor`, `cloud_provider`, `ai_lab`, `enterprise_user`, `integrator_consultant`, or `data_platform_provider`. Whereas **role_label** describes the firm's role within the focal ecosystem, **firm_type** describes the kind of company it is more generally in the broader market environment.

The **direct_competitor** variable records whether a firm overlaps with the platform sponsor in downstream market arenas relevant to the coopetition interpretation used in this study. Importantly, centrality measures are not used to assign any of these categories.

The detailed annotation layer is maintained separately and merged into the firm-level metrics table through a left join on normalized firm names. Firms outside the focal subset remain structurally present in the network but are left unclassified in the detailed annotation fields. For downstream aggregation and visualization, additional coarse grouping fields are derived across the merged analysis file from the available detailed annotations. In particular, **role_group** provides a high-level strategic grouping relative to the platform sponsor, while **direct_competitor_group** provides a grouped competitor-status field. These derived grouping variables are used for summary comparisons and plots. Outside the focal subset, fallback grouping values may still appear in the merged metrics file for processing consistency, but they should not be interpreted as equivalent to the detailed manual annotations.

3.9.3 Competitive Arena Classification and Edge-Type Aggregation

Using the binary **direct_competitor** indicator, collaboration ties are categorized according to the competitor status of the two connected firms. This step does not create a new network. Rather, it assigns a quantitative edge-type label to each observed edge in order to compare collaboration structure across different competitive pairings.

After merging competitor labels into the firm-level node table, each edge is assigned one of four dyadic categories:

1. competitor–competitor
2. competitor–non-competitor
3. non-competitor–non-competitor
4. unknown, if one or both firms do not have a competitor classification

These categories are not taken from a fixed standard classification in prior literature. Instead, they are created for this study to describe the type of competitive relationship between the two firms connected by each edge. This edge typing is a deterministic transformation of the existing firm-level annotations and leaves the underlying graph unchanged. Its purpose is analytical comparison: it makes it possible to examine whether collaboration differs across competitive arenas while preserving the trace-derived network representation.

Subsequent aggregation compares both the number of edges and the cumulative edge weights across the four edge-type categories. These summaries are computed separately for the overall, core-only, and peripheral-only variants. This preserves comparability across repository layers because the annotation logic is applied in parallel to networks that were constructed using the same underlying tie rule.

3.9.4 Scope and Limitations

The detailed annotation framework applies only to the focal subset of firms. As a result, firms outside that subset remain part of the structural network but do not contribute to the finer role-based subgroup interpretation. In addition, competitor status is inferred from publicly observable market orientation and therefore simplifies potentially nuanced, overlapping, or evolving competitive relationships.

More broadly, the annotation layer should be understood as an interpretive overlay rather than a trace-derived measurement stage. It is added after network reconstruction and metric computation, and it does not alter the structural properties of the observed network. This separation preserves transparency between computationally derived outputs and theory-guided manual categorization.

3.10 Validity, Measurement Boundaries, and Robustness Considerations

This section summarizes the main methodological boundaries already discussed throughout the chapter. The collaboration network is reconstructed from shared file-level modification activity on the default branch within the defined observation window. Accordingly, collaboration is operationalized as joint engagement with repository artifacts and should be interpreted as a structural proxy for inter-firm technical co-presence rather than as evidence of direct interpersonal communication, contractual partnership, or formal alliance formation.

Firm attribution is inferred from observable digital signals, with canonical email-domain mapping as the primary signal and GitHub profile metadata as a secondary fallback. Although deterministic rules are applied wherever possible and ambiguous cases are retained in residual or uncertain categories, affiliation inference remains subject to measurement uncertainty. Potential sources of error include outdated or incomplete profile fields, personal or masked email addresses, and domains that do not map cleanly onto organizational boundaries. In addition, affiliations are assigned at contributor level and held constant over the observation window. This supports internal consistency in aggregation, but it may simplify cases in which contributors changed employers during the study period.

The core-peripheral distinction is also an analytical operationalization rather than a direct measure of true architectural importance. It is implemented through deterministic file-path prefix rules that enable transparent comparison across structural parts of the repository. Even so, repository structure does not necessarily correspond perfectly to technological centrality in every case.

3.10.1 Network Construction Assumptions and Analytical Scope

The firm-firm network is built from shared-file co-change within the observation window. This design entails several analytical assumptions. First, co-change may reflect technical interdependence or shared work on the same artifact even when contributions are not temporally synchronized. Second, edge weights capture the breadth of shared file engagement through the number of distinct co-modified files rather than the repeated intensity or technical impact of changes on a single file. Third, ties are modeled as undirected because the network is intended to represent mutual participation in shared artifacts rather than directional influence between firms.

The analysis further focuses on inter-firm coopetition by restricting firm-node construction to contributors classified as `firm_affiliated` and by excluding academic institutions during firm-edge construction through name-based keyword filtering. These boundaries are analytically useful for the present research question, but they may simplify hybrid cases such as research-oriented corporate labs or organizations that combine academic and commercial roles.

3.10.2 Robustness, Reproducibility, and Transparency

Robustness is assessed through the parallel construction of three network variants: the overall network, the core-only network, and the peripheral-only network. These variants are generated using the same tie-construction and metric-computation procedures, so differences in the resulting structural patterns can be interpreted in light of repository scope rather than methodological inconsistency.

More broadly, the workflow is implemented through scripted transformations with fixed configuration parameters. Given the same repository state, branch selection, observation window, and mapping files, the pipeline reproduces the same outputs. This supports transparency and replicability across the full sequence from trace extraction to network measurement. Where needed, numerical fallback procedures are used only to ensure computational stability, for example in the computation of eigenvector centrality, without changing the underlying network definition or the substantive interpretation of the graph.

4 Empirical Analysis

This chapter reports the results of the empirical analysis and addresses the three research questions in sequence. It examines how inter-firm collaboration is structured within the PyTorch ecosystem, how firms are positioned within that structure, and how competitive relationships are reflected in collaboration patterns.

RQ1 examines how the collaborative dimension of inter-firm coopetition can be modeled using repository-based network analysis. The focus is on the structural properties of the reconstructed firm-level networks and the collaboration patterns visible within them.

RQ2 examines firms' positions within these networks and considers whether differences in collaboration intensity, embeddedness, and brokerage align with distinct strategic roles in a vendor-led open-source ecosystem.

RQ3 focuses on the competitive dimension by comparing collaboration patterns between firms that are direct market competitors and those that are not. The analysis assesses whether rivalry is associated with systematic differences in collaboration intensity or tie structure.

Section 4.1 analyzes the structural characteristics of the overall, core-only, and peripheral-only collaboration networks in order to address (RQ1). Section 4.2 interprets firms' network positions in relation to the strategic annotation framework (RQ2). Section 4.3 applies a competitor lens to the same networks to examine differences across competitive pairings (RQ3).

4.1 RQ1 Modeling Inter-Firm Collaboration from Repository Traces

RQ1: How can repository mining and social network analysis be used to model the collaborative dimension of inter-firm coopetition in open-source ecosystems?

4.1.1 Empirical Basis and Scope Differentiation

Figure 4-1 summarizes the empirical scale of repository activity and its distribution across the two scope variants used in the analysis. Over the observation window from 2014-06-01 to 2025-06-30, the PyTorch repository records 5,087 stabilized contributors and 88,698 unique commits. These commits correspond to 447,495 commit-file modification pairs, which provide the basis for identifying shared artifact engagement between firms.

Measure	# Count
Contributors (after identity stabilization)	5,087
Unique commits	88,698
Commit-file pairs (overall)	447,495
Commit-file pairs (core scope)	191,661
Commit-file pairs (peripheral scope)	255,834

Figure 4-1: Dataset overview and scope distribution

As Figure 4-1 shows, repository activity is distributed across both core and peripheral areas of the repository. Of the 447,495 commit-file pairs, 191,661 fall within the defined core scope, while 255,834 are associated with peripheral components. In relative terms, approximately 43% of observable file-level engagement occurs in core subsystems and 57% in peripheral areas.

This distribution shows that development activity is not confined to the core scope. A substantial share of file-level engagement occurs in peripheral areas, while the absolute volume of core activity also remains high. The core scope alone contains nearly two hundred thousand file-level modification events, which provides a sufficient empirical basis to analyze the core and peripheral network variants separately. The presence of substantial activity in both scopes supports the parallel construction of overall, core-only, and peripheral-only collaboration networks. The distinction between scopes is therefore not applied to marginal activity, but reflects clear differences in where development work is concentrated within the repository.

4.1.2 Structural Properties of the Overall Collaboration Network

Figure 4-2 reports the highest-weight inter-firm ties in the overall collaboration network and already indicates a highly uneven distribution of inter-firm engagement. Rather than showing a flat structure with evenly distributed ties, the strongest connections are concentrated among a limited set of actors.

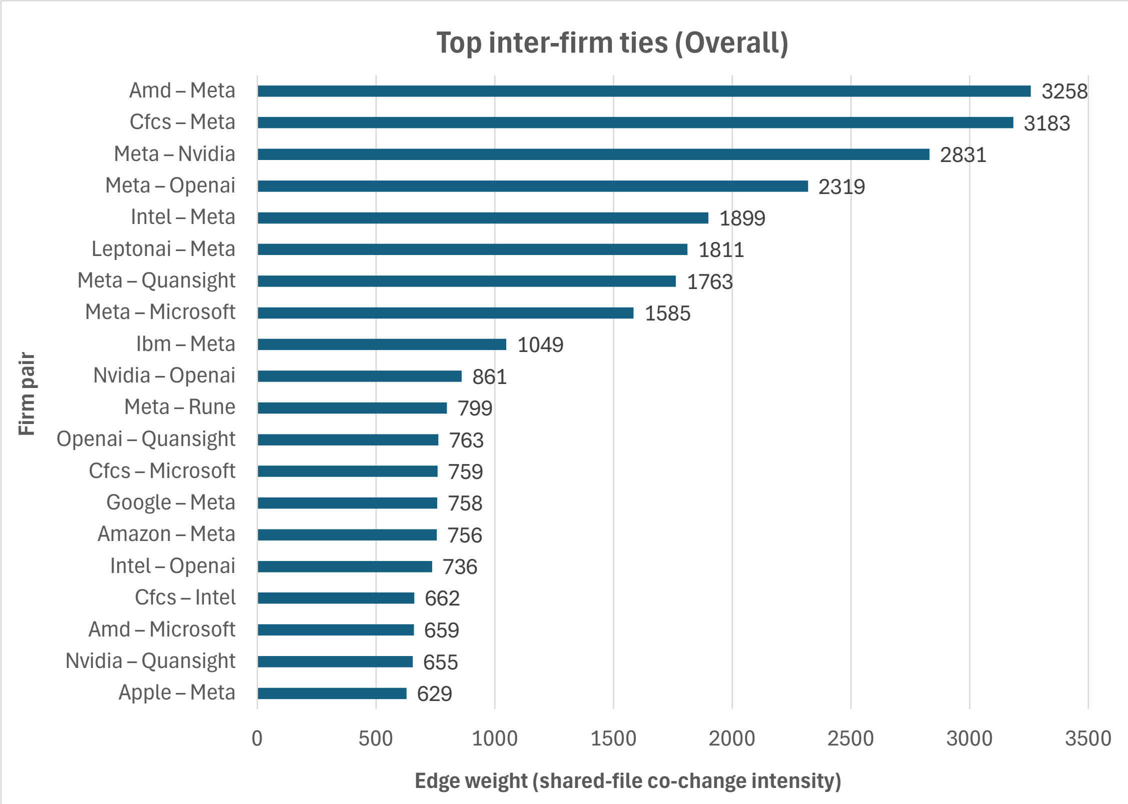


Figure 4-2: Top inter-firm collaboration ties (overall)

The most intensive shared-file relationships are dominated by pairings involving the platform sponsor. The strongest tie in the network connects AMD and Meta with an edge weight of 3,258 distinct shared files. This is followed closely by CFCs–Meta (3,183) and Meta–Nvidia (2,831). Several additional high-weight ties, including Meta–OpenAI (2,319) and Intel–Meta (1,899), further reinforce the central position of Meta in the collaboration structure. A clear pattern emerges: the majority of the highest-ranked ties involve the same focal firm. This indicates that shared artifact engagement in the repository is not evenly distributed across the ecosystem. Instead, a small number of firms repeatedly co-modify the same files with the platform sponsor, creating disproportionately strong ties in the network. Beyond this upper tier, edge weights decline steadily. While there remains a visible cluster of moderately strong ties, such as Nvidia–OpenAI (861) and OpenAI–Quansight (763), the drop from the top-ranked pairs to the middle of the distribution is substantial. The difference between the strongest tie (3,258) and ties ranked around twentieth place (approximately 600-700) highlights the sharp decline in collaboration intensity.

Two structural features are evident. First, collaboration intensity is concentrated among a limited number of firm pairs. Second, the strongest ties cluster around a small set of central actors, especially the platform sponsor and major hardware and AI firms. This is consistent with a vendor-led ecosystem in which shared development activity centers on a coordinating firm and a group of highly engaged complementors and competitors. These ties are derived from shared-file co-modification, so they reflect repeated joint engagement with the same repository artifacts rather than formal partnerships. Overall, the PyTorch ecosystem displays visible but unevenly distributed inter-firm collaboration.

4.1.3 Scope-Specific Collaboration Structures (Core vs. Peripheral)

Figures 4-3 and 4-4 compare the highest-weight ties in the core and peripheral network variants. Separating the analysis by repository scope reveals clearer differences in how collaboration is organized across these two parts of the repository.

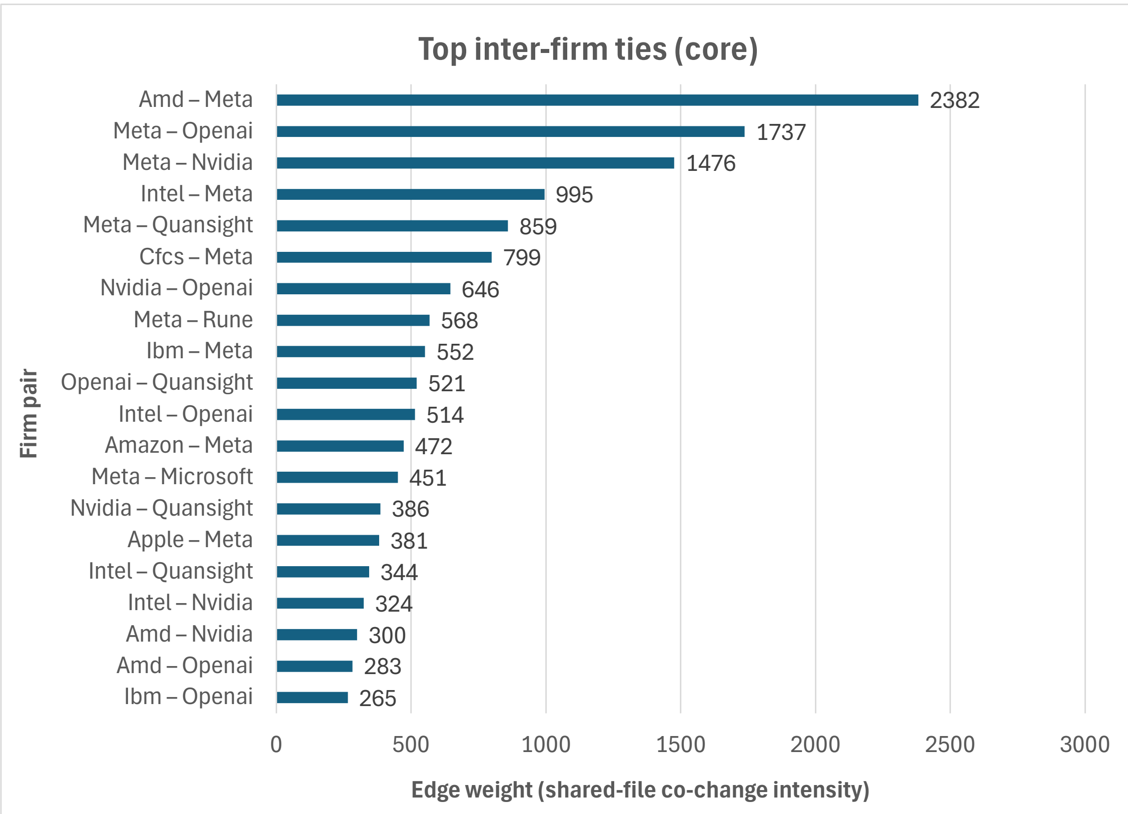


Figure 4-3: Top inter-firm collaboration ties (core)

As shown in Figure 4-3, the core network displays a strongly hierarchical structure. The most intensive tie in the core layer connects AMD and Meta with an edge weight of 2,382 distinct shared files. This is followed by Meta–OpenAI (1,737) and Meta–Nvidia (1,476). Several additional ties involving Meta, such as Intel–Meta (995) and Meta–Quansight (859), remain substantially stronger than the rest of the distribution. Two features are particularly evident. First, the strongest core ties are consistently anchored around the platform sponsor. Second, the decline from the top-ranked ties to the lower half of the top-20 list is steep. By the time the ranking reaches positions fifteen to twenty, edge weights fall below 400. This suggests that shared engagement within the core scope is concentrated among a relatively small group of firms.

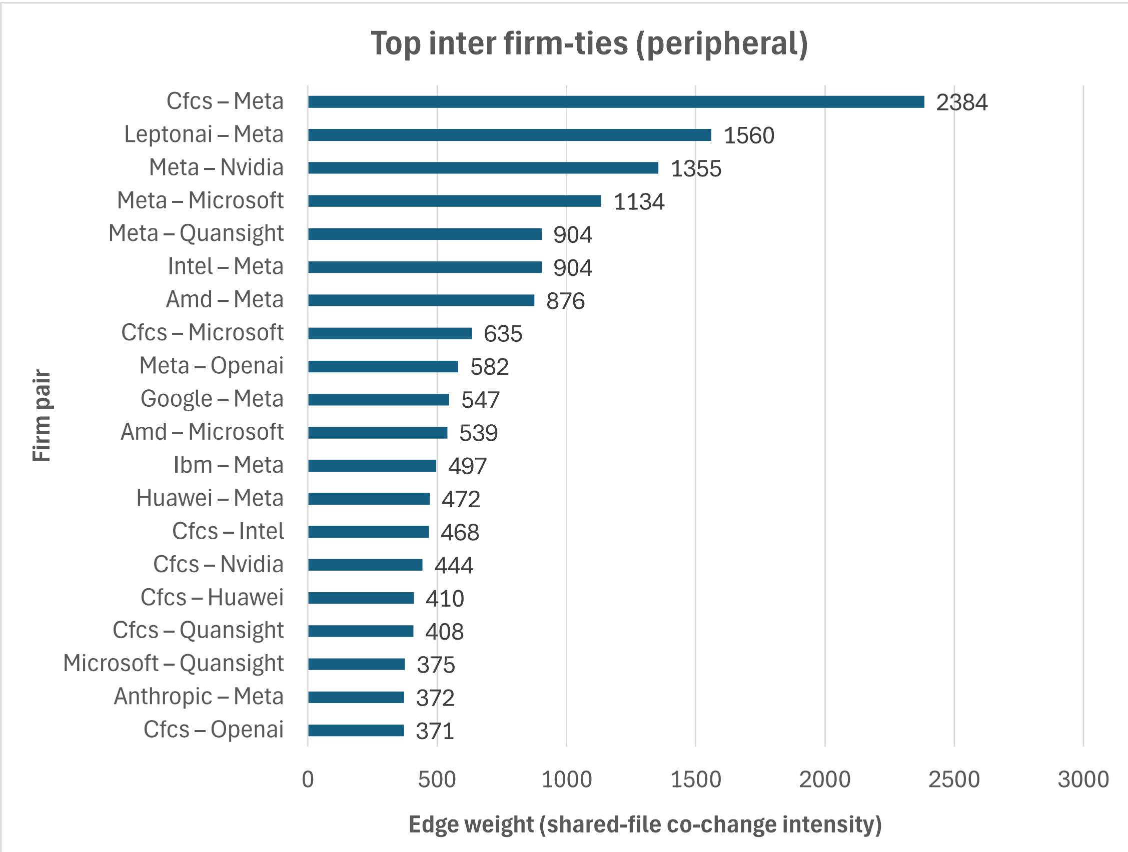


Figure 4-4: Top inter-firm collaboration ties (peripheral)

Figure 4-4 shows a somewhat different pattern for the peripheral network. In the peripheral layer, the strongest tie connects CFCs and Meta with an edge weight of 2,384, followed by LeptonAI–Meta (1,560) and Meta–Nvidia (1,355). Although Meta remains prominently involved in the highest-ranked ties, the distribution beyond the top few pairs appears less steep than in the core network. A larger number of firm pairs cluster in a moderate edge-weight range, indicating a broader spread of mid-intensity ties.

Compared to the core network, the peripheral network shows a more distributed structure among secondary actors. While the platform sponsor continues to appear in many of the strongest ties, other firm pairings not directly involving Meta also occupy visible positions in the top-20 list. This suggests that peripheral components provide a broader surface for interaction among firms beyond the most concentrated set of core ties.

Figure 4-5 complements the tie-level comparison by showing the ranked firm-level distribution of weighted degree in the core network on a logarithmic scale.

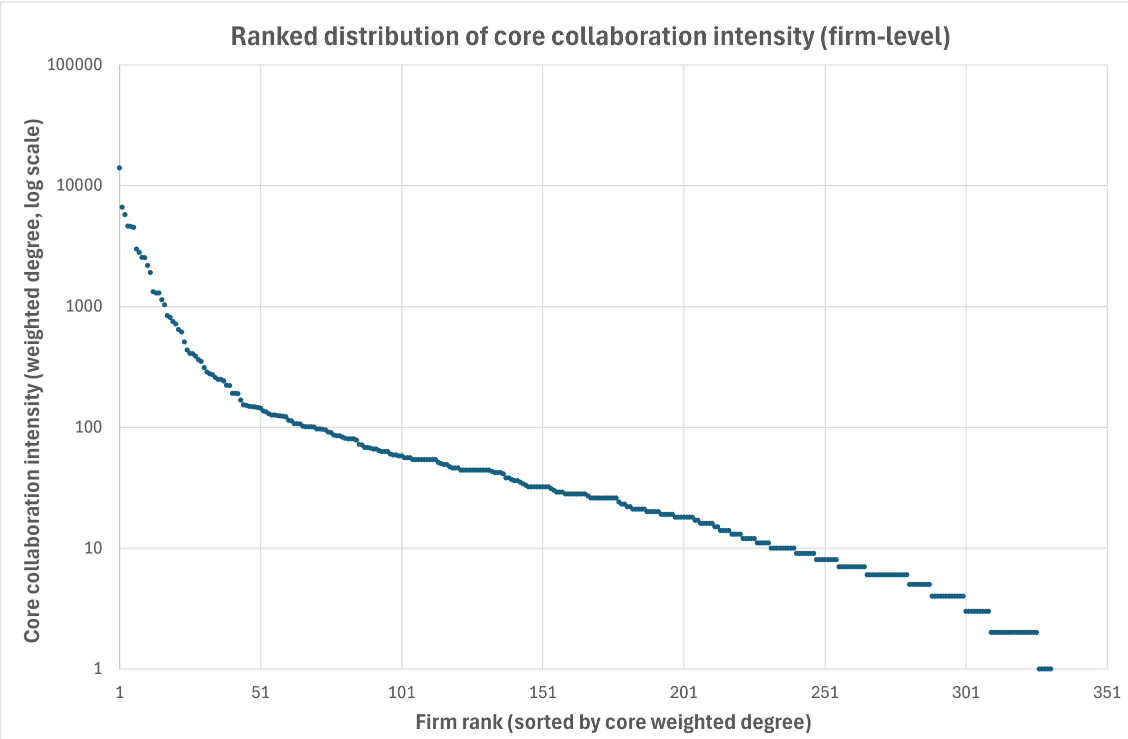


Figure 4-5: Ranked distribution of core collaboration intensity (firm-level, log scale)

The firm-level distribution of collaboration intensity within the core network further reinforces this structural difference. Figure 4-5 plots firms in descending order of weighted degree within the core network on a logarithmic scale. The steep initial decline from the highest-ranked firms to the rest of the distribution indicates pronounced structural skewness. A small number of firms accumulate substantially higher levels of shared-file engagement in the core scope, while the majority of firms exhibit comparatively low collaboration intensity. After the highest-ranked firms, weighted-degree values decline sharply and then taper gradually across a long tail of lower-intensity actors. This heavy-tailed pattern confirms that the concentration observed at the tie level in Figure 4-3 (p. 52) extends to the broader firm-level structure of the core network. Core collaboration is therefore not only characterized by a few dominant firm pairs but also by a hierarchical distribution of overall engagement across firms.

Overall, the scope comparison points to a clear structural difference. Core collaboration is more tightly concentrated around a small set of highly engaged firms and shows sharper hierarchical differentiation at both tie and firm level. Peripheral collaboration, although still structured and unequal, is distributed more broadly across firms. The repository-based analysis therefore captures not only the concentration of collaboration, but also how that concentration changes between the core and peripheral network variants.

4.1.4 Summary of RQ1

The analyses in this section show that repository-based network reconstruction provides a useful structural representation of inter-firm collaboration within the PyTorch ecosystem. The empirical scale of activity offers a substantial basis for modeling collaboration, and the volume of commit-file modification events across both core and peripheral areas indicates that inter-firm engagement is neither marginal nor isolated.

The overall network shows that shared-file engagement is concentrated among a limited number of firm pairs, especially around the platform sponsor and major hardware and AI actors. The steep decline in edge weights beyond the top-ranked pairs further indicates that collaboration intensity is unevenly distributed across the ecosystem. The scope-specific analysis also shows that this concentration is not the same across repository layers. Core subsystems exhibit sharper hierarchical differentiation, both at the level of individual ties and in the firm-level distribution of collaboration intensity, whereas peripheral collaboration shows a broader distribution of moderate-intensity ties.

These findings address RQ1 by showing that repository mining combined with social network analysis can capture concentration, hierarchy, and scope-specific variation in inter-firm engagement. The reconstructed networks therefore provide an empirical representation of the collaborative dimension of inter-firm coopetition in the PyTorch ecosystem.

4.2 RQ2 Structural Positions and Strategic Roles

RQ2: How do firms' structural positions and collaboration patterns within the PyTorch project reflect their strategic roles in a vendor-led coopetitive ecosystem?

The previous section described the overall structure of inter-firm collaboration. This section turns to interpreting that structure in relation to ecosystem roles derived from the manual annotation framework introduced in Chapter 3. The analyses in this section are based on a focal subset of 54 firms. These firms were identified through a weighted-degree ranking in the overall network and selected for detailed annotation where a sufficiently clear organizational identity and publicly observable market position could be established. They represent the firms with the highest levels of collaboration intensity and the most central positions in the network.

All structural metrics, including weighted degree, eigenvector centrality, and betweenness centrality, were computed on the full firm-level network. For the interpretive analyses presented below, these metric values are filtered to the annotated subset. The annotation layer serves purely as an interpretive overlay and does not alter the network structure or the way metrics are computed. Unless otherwise stated, the comparisons in this section are based on the overall firm-level network. The scope distinction introduced in RQ1 is taken up later in this section through the core-based strategy map and the backbone visualization.

Each firm in the focal subset is assigned three manually coded variables: *role_label*, *firm_type*, and *direct_competitor*. From these detailed annotations, derived grouping variables, such as *role_group* and *direct_competitor_group*, are used for aggregation and visualization. Here, *direct_competitor* is a binary indicator of downstream market overlap with the platform sponsor in market arenas relevant to the coopetition interpretation used in this study. Firms outside the focal subset remain structurally present in the network but are not included in the fine-grained role-based subgroup comparisons.

4.2.1 Collaboration Intensity Across Role Groups

To examine whether collaboration intensity differs across broad ecosystem roles, weighted-degree values are aggregated by the derived *role_group* variable within the annotated subset. Weighted degree is computed on the overall firm level network, and the comparison presented here shows the average collaboration intensity across role group categories among the annotated firms. Figure 4-6 reports the mean collaboration intensity across the three role groups.

The platform sponsor is structurally distinct from the two non-sponsor groups. Because the sponsor's value is much larger than the values for competitors and other firms, the difference between the two non-sponsor groups is less visible in that full comparison. Figure 4-7 (p. 58) therefore shows the non-sponsor comparison separately.

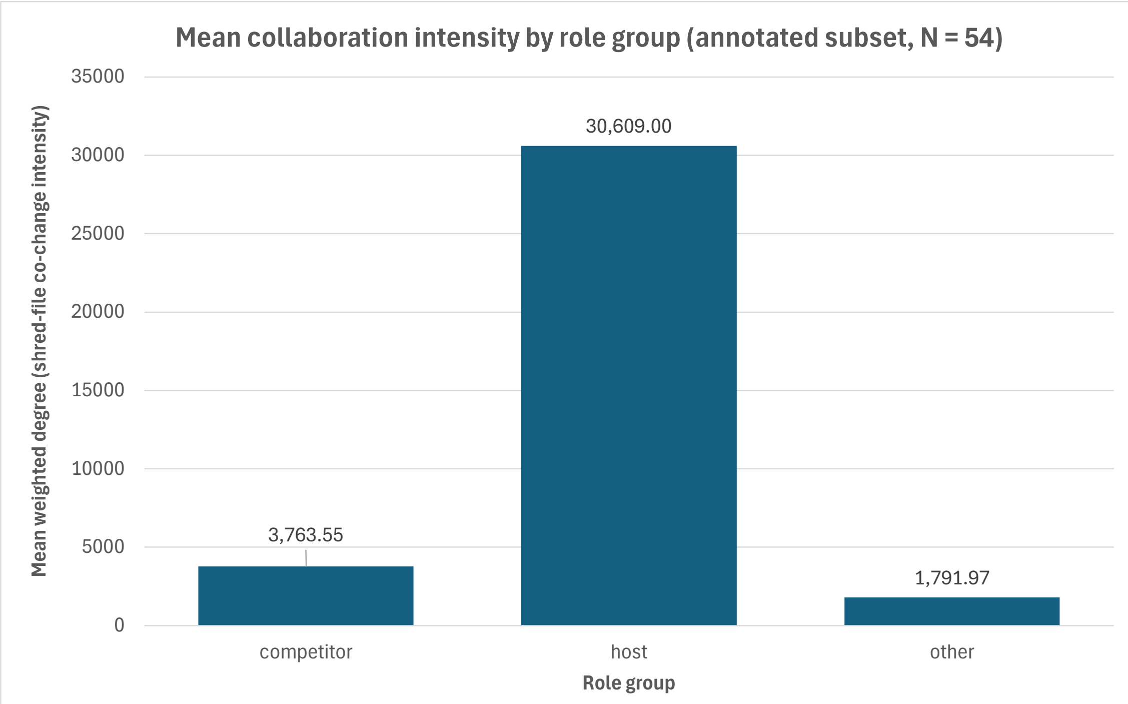


Figure 4-6: Mean collaboration intensity by role group for the annotated subset.

As shown in Figure 4-6, collaboration intensity differs markedly across the three role groups. The ecosystem orchestrator (platform sponsor) exhibits by far the highest mean weighted degree, with a value of 30,609. This substantially exceeds the mean collaboration intensity of direct competitors (3,763.55) and other firms (1,791.97). The size of this gap highlights the structurally dominant position of the sponsor within the collaboration network. On average, the sponsor's shared-file engagement is an order of magnitude greater than that of competitor firms and even more distant from firms classified in the "other" category. This confirms that the sponsor occupies a uniquely central position in terms of the range of its collaborative ties.

At the same time, Figure 4-7 makes clear that firms coded as direct competitors of the sponsor exhibit a noticeably higher mean collaboration intensity than firms in the "other" category. This suggests that firms with downstream market overlap with the sponsor are not marginal actors within the ecosystem. Instead, they form a secondary tier of structurally engaged participants with substantial shared technical activity.

Overall, the results indicate a clear stratification of collaboration intensity across broad ecosystem roles. Structural prominence in the PyTorch collaboration network is highest

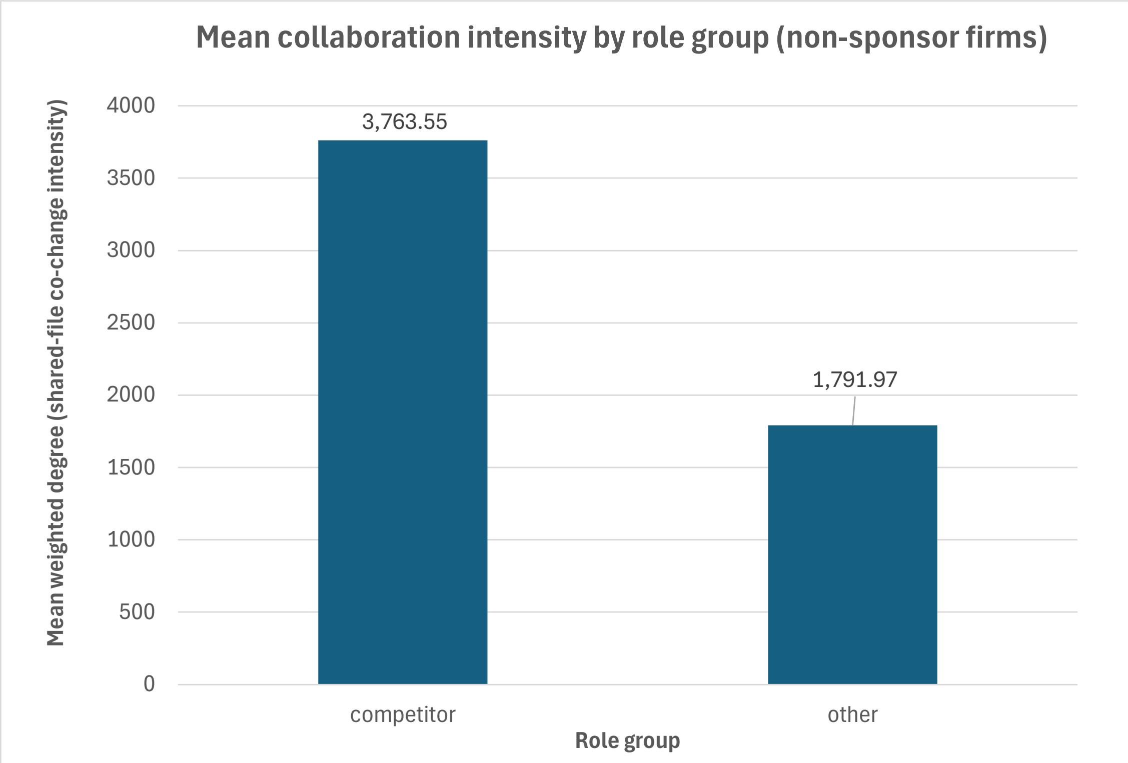


Figure 4-7: Mean collaboration intensity for non-sponsor role groups (annotated subset)

for the platform sponsor, lower for direct competitors, and lowest for the remaining firms in the annotated subset.

4.2.2 Collaboration Intensity by Fine-Grained Role Labels

The detailed *role_label* categories used in the following analysis correspond directly to the fine-grained manual annotations described in Chapter 3. These labels were assigned to the focal subset of 54 firms on the basis of publicly observable organizational identity, market positioning, and ecosystem role within the PyTorch setting. Chapter 3 introduced a smaller set of broader role families only as a conceptual framework for explaining the annotation logic. By contrast, the present section uses the full set of fine-grained labels recorded in the annotation file for the focal subset. A complete listing of annotated firms and their assigned role labels is provided in Appendix A Table 1-1 for transparency. To examine variation within and across these detailed categories, weighted-degree values are analyzed by *role_label* for the annotated subset. As in the previous subsection, weighted degree is computed on the full overall network, and the values presented here reflect mean collaboration intensity within each fine-grained role category among the annotated firms. Figure 4-8 reports mean collaboration intensity across the detailed role labels.

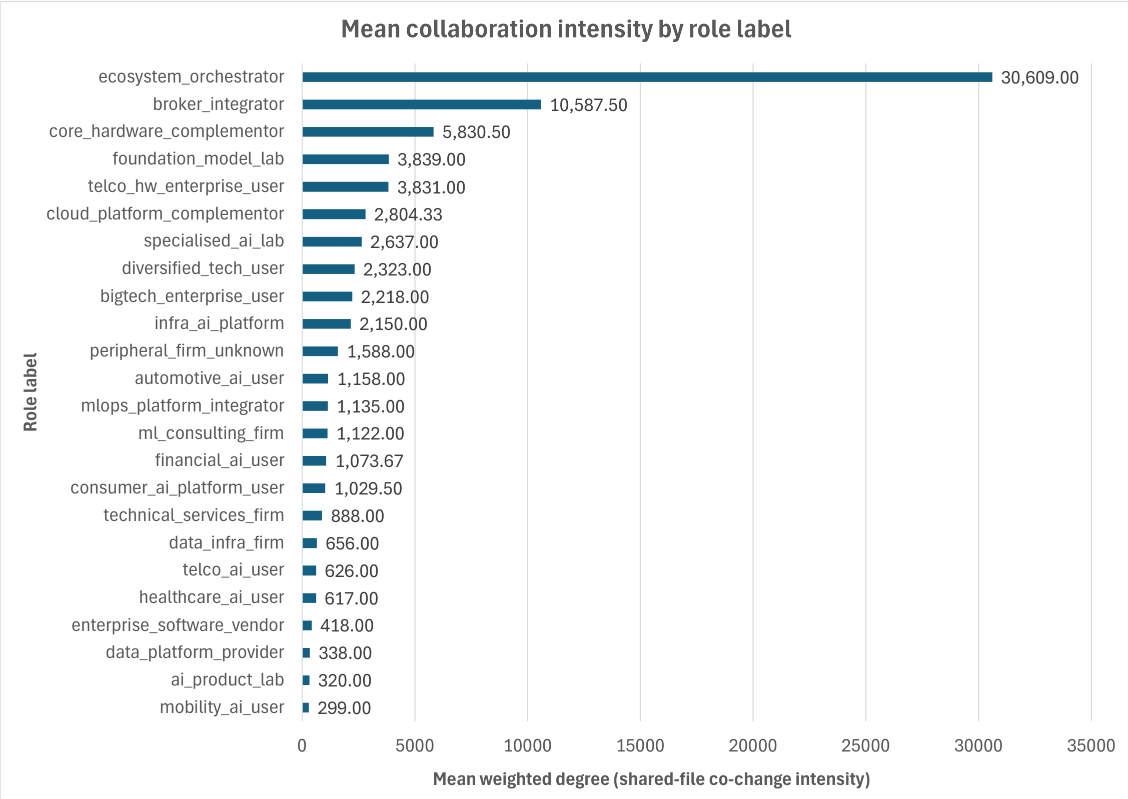


Figure 4-8: Mean collaboration intensity by fine-grained role label (annotated subset)

Clear differences are visible in collaboration intensity across detailed ecosystem roles. The ecosystem orchestrator stands apart from all other categories, with a mean weighted degree of 30,609. This confirms the sponsor's exceptional breadth of shared-file engagement and reinforces its dominant structural position within the collaboration network. Beyond the sponsor, a second group of highly engaged actors becomes visible. The *broker-integrator* category exhibits a comparatively high mean collaboration intensity (10,587.50), followed by *core hardware complementors* (5,830.50). These roles are associated with infrastructural integration and hardware-level complementarity, which is consistent with their elevated levels of shared engagement in the repository.

A further cluster including *foundation model labs*, *telco hw enterprise user*, and *cloud platform complementors* shows moderate collaboration intensity, generally ranging between approximately 2,500 and 4,000. These actors are clearly embedded in the ecosystem but do not reach the level of engagement observed among infrastructure-oriented complementors and integration-focused firms.

At the lower end of the distribution, several specialized or domain-specific categories, such as *enterprise software vendors*, *data platform providers*, *AI product labs*, and *mobility AI users*, display comparatively limited average collaboration intensity. These firms remain part of the structural network but exhibit narrower shared-file engagement relative to the upper tiers.

Overall, the fine-grained breakdown reveals a layered hierarchy that extends beyond the simple distinction between sponsor, competitors, and others. Structural prominence appears closely associated with infrastructural proximity and integration capacity, whereas more downstream actors occupy comparatively less collaboration-intensive positions. These differences suggest that network position closely reflects firms' ecosystem roles.

4.2.3 Collaboration Intensity by Firm Type

To further assess whether collaboration intensity aligns with firms' broader market identities, weighted-degree values are compared across the *firm_type* categories assigned to the annotated subset. As in the previous analyses, weighted degree is computed on the full overall network, and the values presented here reflect mean collaboration intensity within each firm-type category among the annotated firms. Figure 4-9 (p. 61) reports mean collaboration intensity across the firm-type categories.

A substantial variation in collaboration intensity across firm types is visible in Figure 4-9. The platform sponsor occupies a clearly dominant position, with a mean weighted degree of 30,609, far exceeding all other categories. This further highlights the sponsor's central role within the ecosystem and its extensive shared-file engagement across the network.

Among non-sponsor firms, GPU vendors exhibit the highest average collaboration intensity (5,830.50), followed by AI labs (3,052.17) and cloud providers (2,950.20). These firm types are closely associated with compute infrastructure, model development, and platform-level capabilities, which corresponds to their comparatively elevated structural engagement in the repository. Integrator and consulting firms show moderate collaboration intensity (2,323.58), suggesting meaningful but less dominant participation in shared development activities. The residual "other" category (1,588.00) and enterprise users (1,409.14) occupy the lower end of the distribution, indicating more selective or application-oriented involvement in the framework. Overall, the firm-type comparison complements the role-based analysis by showing that structural prominence is strongly

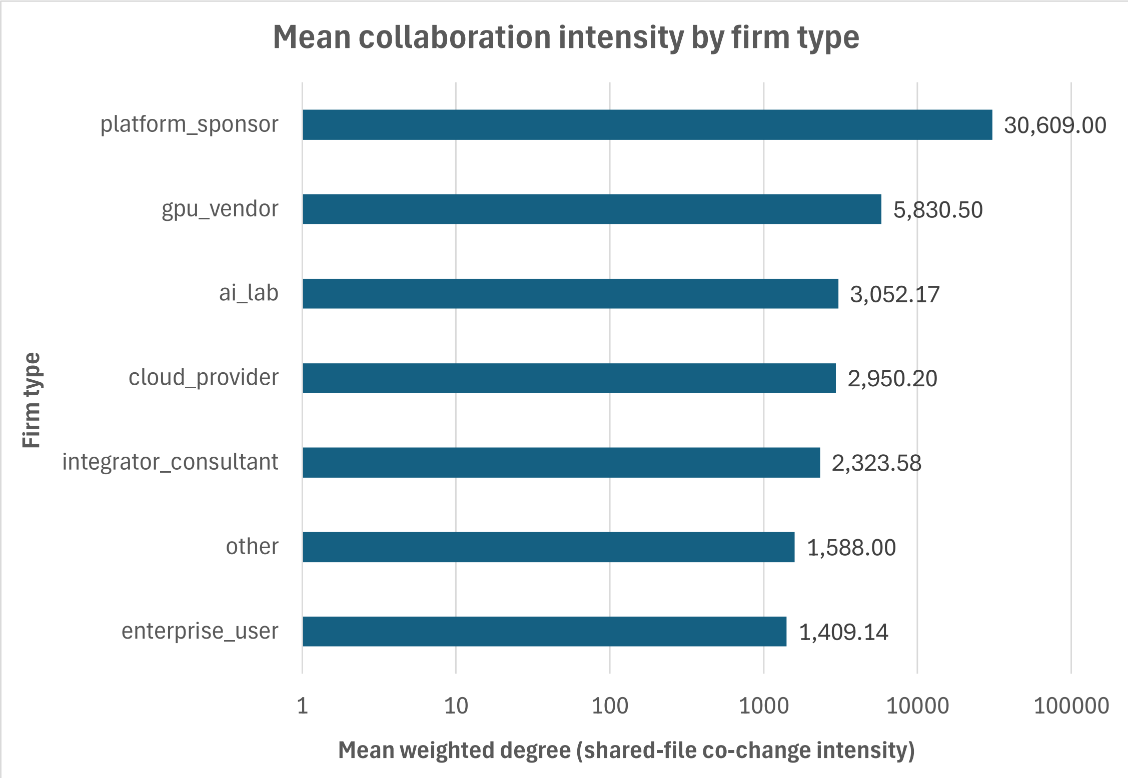


Figure 4-9: Mean collaboration intensity by firm type (annotated subset, log scale)

associated with infrastructural and compute-oriented market identities. Firms positioned closer to the technological core of the AI stack tend to exhibit higher collaboration intensity, whereas more downstream and application-oriented actors occupy comparatively less collaboration-intensive positions within the shared development structure.

4.2.4 Embeddedness and Brokerage as Distinct Structural Dimensions

Collaboration intensity captures the breadth of shared-file engagement, but it does not fully describe how firms are positioned within the broader collaboration structure. To better understand firms' positions within the network, this subsection examines embeddedness, measured through eigenvector centrality, and brokerage, measured through betweenness centrality. As in previous analyses, both metrics are computed on the full overall network and then filtered to the annotated subset for comparison across firm types. Figure 4-10 reports mean eigenvector centrality across firm types.

Clear differences can be seen in embeddedness across firm types. The platform sponsor exhibits by far the highest mean eigenvector centrality (0.5953), substantially exceeding

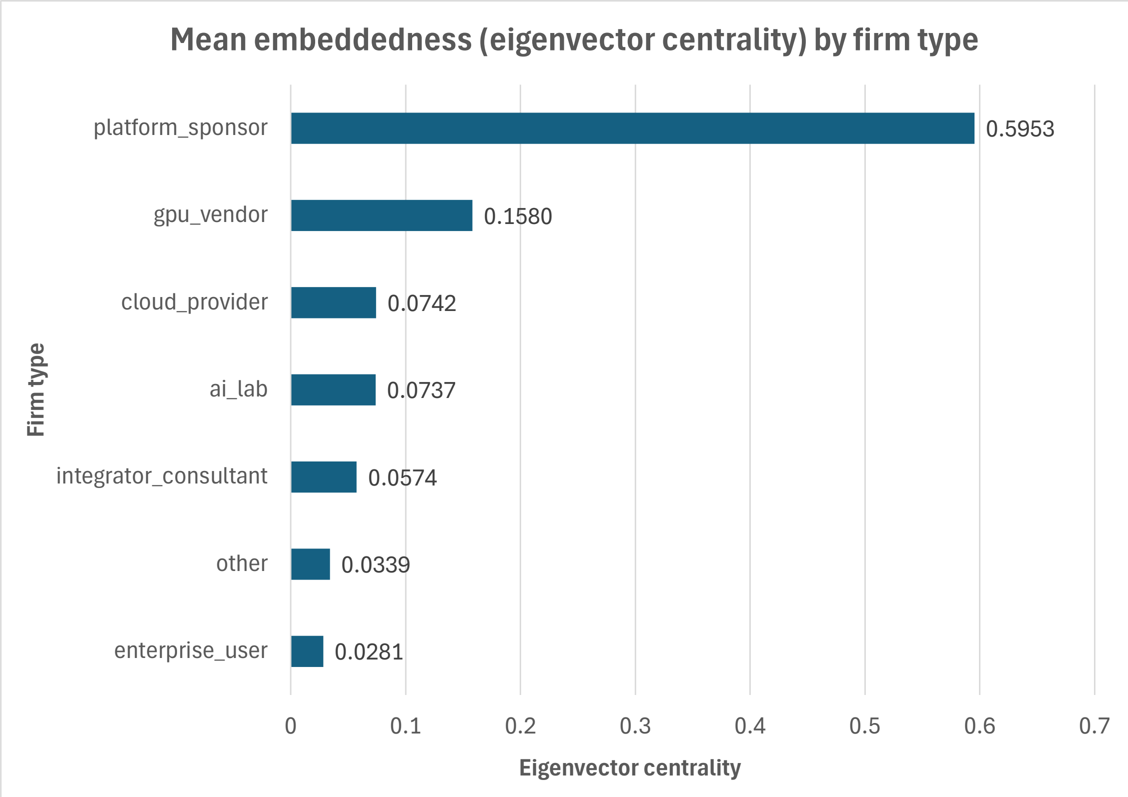


Figure 4-10: Mean eigenvector centrality by firm type for annotated subset.

all other categories. This indicates not only high collaboration intensity, but also strong ties to other central firms in the network.

Among non-sponsor firms, GPU vendors display the highest embeddedness (0.1580), while cloud providers (0.0742) and AI labs (0.0737) occupy a secondary tier. These values suggest that infrastructure-oriented and compute-centric firms are more deeply integrated into the central collaboration core of the ecosystem. Integrator-consultants (0.0574) also show meaningful embeddedness, while firms in the residual "other" category (0.0339) and enterprise users (0.0281) exhibit comparatively lower values, indicating that although they participate in collaboration, they are less connected to the most central actors. Figure 4-11 reports mean betweenness centrality across firm types.

Here, brokerage/betweenness follows a different pattern from the one suggested by eigenvector centrality (Figure 4-10). The platform sponsor exhibits the highest mean betweenness centrality by a wide margin (0.2192), followed by GPU vendors (0.0268), cloud providers (0.0154), and AI labs (0.0139). Integrator-consultants (0.0125), firms in the residual "other" category (0.0066), and enterprise users (0.0061) occupy lower positions.

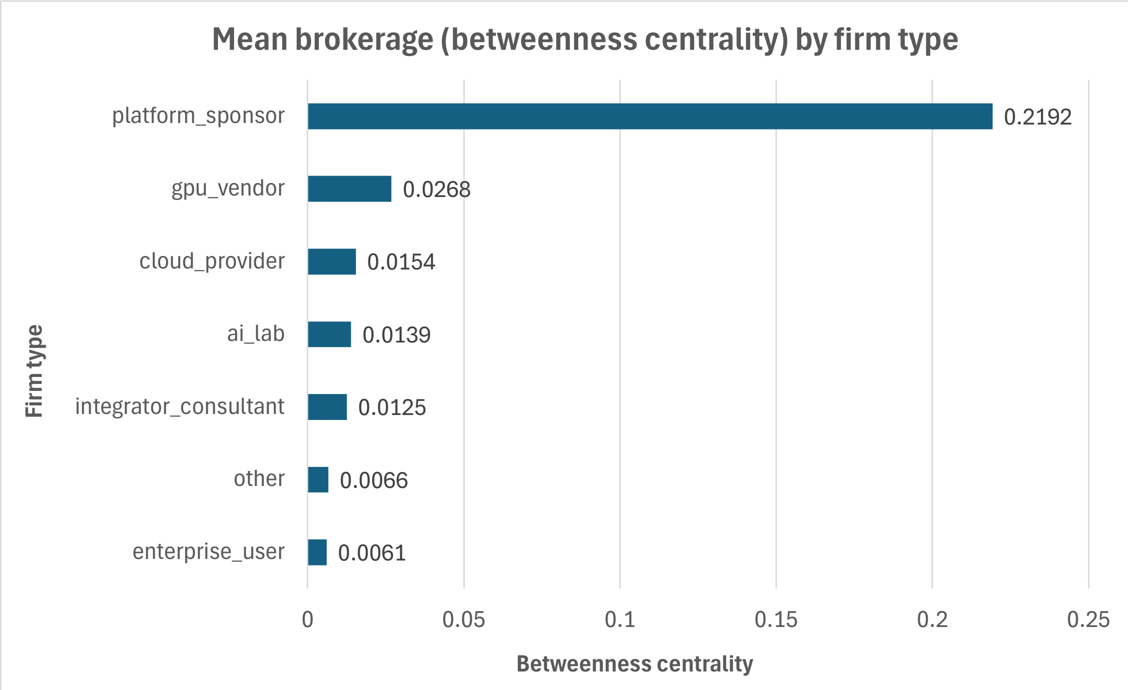


Figure 4-11: Mean betweenness centrality by firm type for annotated subset.

This contrast between embeddedness and brokerage is informative. The sponsor is not only the most collaboration-intensive and most embedded actor, but also the firm that occupies the strongest intermediary position in these firm-type averages. GPU vendors, cloud providers, and AI labs also combine non-trivial brokerage with comparatively strong structural integration, whereas enterprise users and the residual "other" category exhibit lower brokerage on average.

Taken together, these results highlight that collaboration intensity, embeddedness, and brokerage represent distinct structural dimensions, but they do not separate firm types in exactly the same way. Firms closest to the technological core of the ecosystem, such as the sponsor and GPU vendors, are deeply embedded within the central collaboration structure and also occupy stronger brokerage positions than the more downstream categories.

4.2.5 Integrated Structural Positioning and Scope Sensitivity

The preceding analyses examined collaboration intensity, embeddedness, and brokerage separately. Taken together, these dimensions offer a more integrated interpretation of firms' structural positioning within the PyTorch ecosystem. Figure 4-12 maps core collaboration intensity (weighted degree) against brokerage (betweenness centrality) at the firm level. The x-axis is shown on a logarithmic scale, which makes it easier to compare firms across the strongly skewed distribution of core collaboration intensity while still preserving the relative spread among lower-intensity actors.

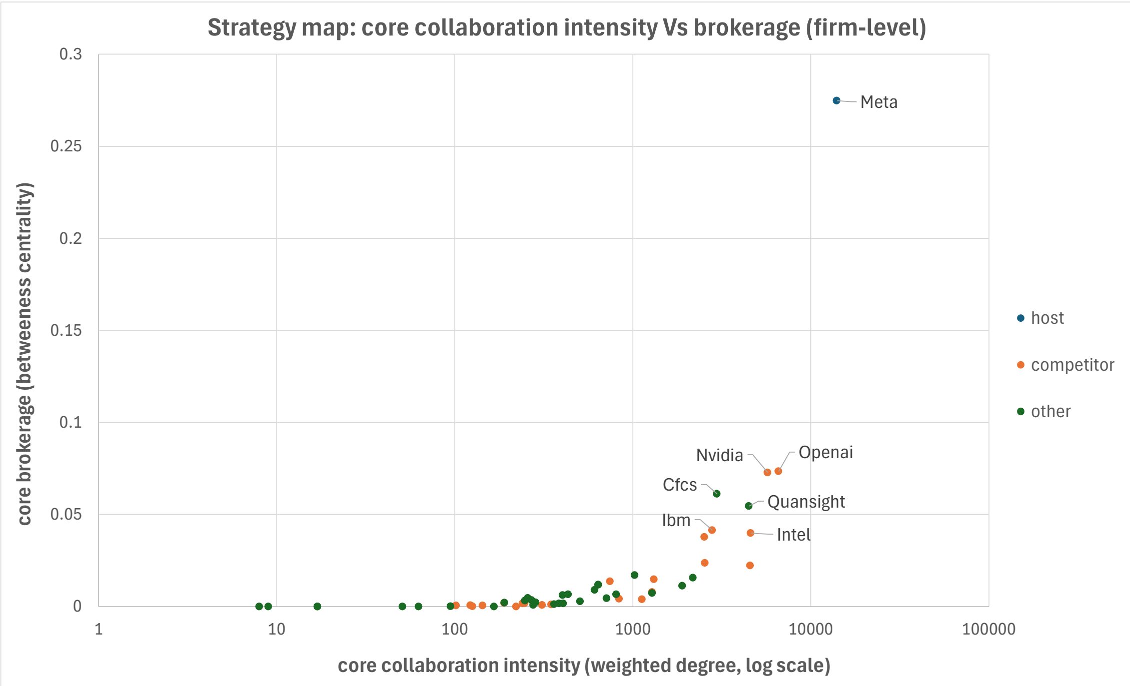


Figure 4-12: Core collaboration intensity vs. brokerage for annotated firms.

Figure 4-12 shows that the platform sponsor (Meta) occupies the most extreme position in the map. It combines by far the highest level of core collaboration intensity with the highest brokerage value among the annotated firms. This indicates that, within the core network, the sponsor is not only the dominant hub of shared-file engagement but also the strongest intermediary actor in the plotted firm-level distribution.

Among the non-sponsor firms, several infrastructure-oriented complementors and selected competitors combine comparatively high core collaboration intensity with visibly elevated brokerage. Firms such as Nvidia and OpenAI lie in the upper portion of the non-host distribution, while Intel, IBM, CFCs, and Quansight also remain clearly above the dense cluster of lower-intensity and lower-brokerage firms. By contrast, many other annotated actors are concentrated near the bottom of the plot, with much lower brokerage values and substantially lower levels of core collaboration intensity. This pattern suggests that structural positioning within the core network is strongly layered. A very small number of firms combine deep engagement in core development with relatively prominent intermediary roles, while most firms remain concentrated in the lower-intensity, lower-brokerage part of the distribution.

Figure 4-13 shows a ranking-based top-50 view of the core network for structural comparison and is separate from the annotated subset used in the role-based analyses.

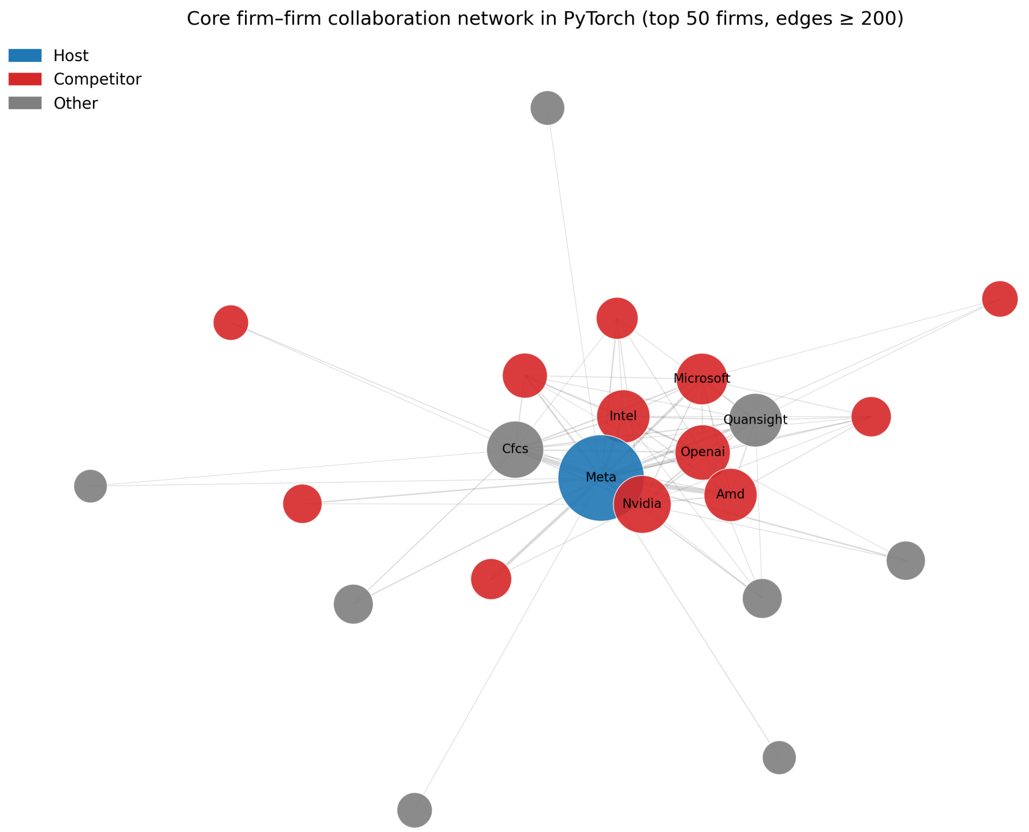


Figure 4-13: High-intensity core firm-firm network (top 50 firms, edges $\geq$ 200).

The high-intensity layer of the core network is concentrated around the sponsor and a relatively small subset of infrastructural complementors and competitors. The reduced number of visible nodes does not indicate data loss. Rather, it reflects the empirical concentration of high-intensity shared-file co-modification among a limited group of actors. Firms that do not maintain ties above this threshold do not appear in the backbone visualization, which further underscores the hierarchical concentration already observed in the metric-based comparisons. Importantly, this visualization complements the earlier metric-based analyses rather than substituting for them. The centrality comparisons and the strategy map summarize firms' structural positions in the complete firm-level network, whereas the backbone visualization (Figure 4-13) shows which high-intensity ties remain visible once attention is restricted to the strongest layer of shared-file co-modification. In this way, the figure does not provide a new metric. Rather, it makes the concentration of strong ties around a limited subset of firms visually explicit. Although the visualizations in this subsection focus on the core network, they are used here to examine scope sensitivity rather than to replace the earlier overall-network comparisons.

Overall, the integrated analysis indicates that firms' structural positions within the PyTorch collaboration network reflect their strategic roles in a vendor-led coopetitive ecosystem. Collaboration intensity captures the breadth of technical engagement, embeddedness reflects integration within the collaboration core, and brokerage highlights intermediary positioning. Considered together, these dimensions point to a layered rather than uniform collaboration structure.

4.2.6 Summary of RQ2

RQ2 asked how firms' structural positions within the PyTorch project reflect the annotated ecosystem roles applied to the focal subset in Chapter 3. The empirical results show that differences in network position align closely with these role distinctions.

Collaboration intensity reveals a clear hierarchy. The platform sponsor occupies the dominant structural position in terms of weighted degree, infrastructure-oriented complementors and selected competitors form a second tier of highly engaged actors, and more downstream firms exhibit more moderate levels of collaboration.

Embeddedness and brokerage further differentiate these roles. Eigenvector centrality indicates that infrastructure-oriented firms are more deeply integrated into the collaboration core, while the brokerage results show that the sponsor also occupies the strongest intermediary position, with GPU vendors, cloud providers, and AI labs following at some distance. The strategy map (Figure 4-12) (p. 64) supports this pattern by showing that only a small number of firms combine high core collaboration intensity with visibly elevated brokerage, whereas most firms remain concentrated in the lower-intensity and lower-brokerage part of the distribution. The filtered high-intensity backbone network (Figure 4-13) (p. 65) points in the same direction by showing that strong shared-file co-modification ties are concentrated among a limited subset of actors. Overall, the findings indicate that structural network position is closely linked to the annotated role structure used in this study and that the collaboration structure of the PyTorch ecosystem is clearly layered rather than uniform.

4.3 RQ3 Competitor Status and Coopetition Patterns

RQ3: How do collaboration patterns differ between firms that are direct market competitors and those that are not?

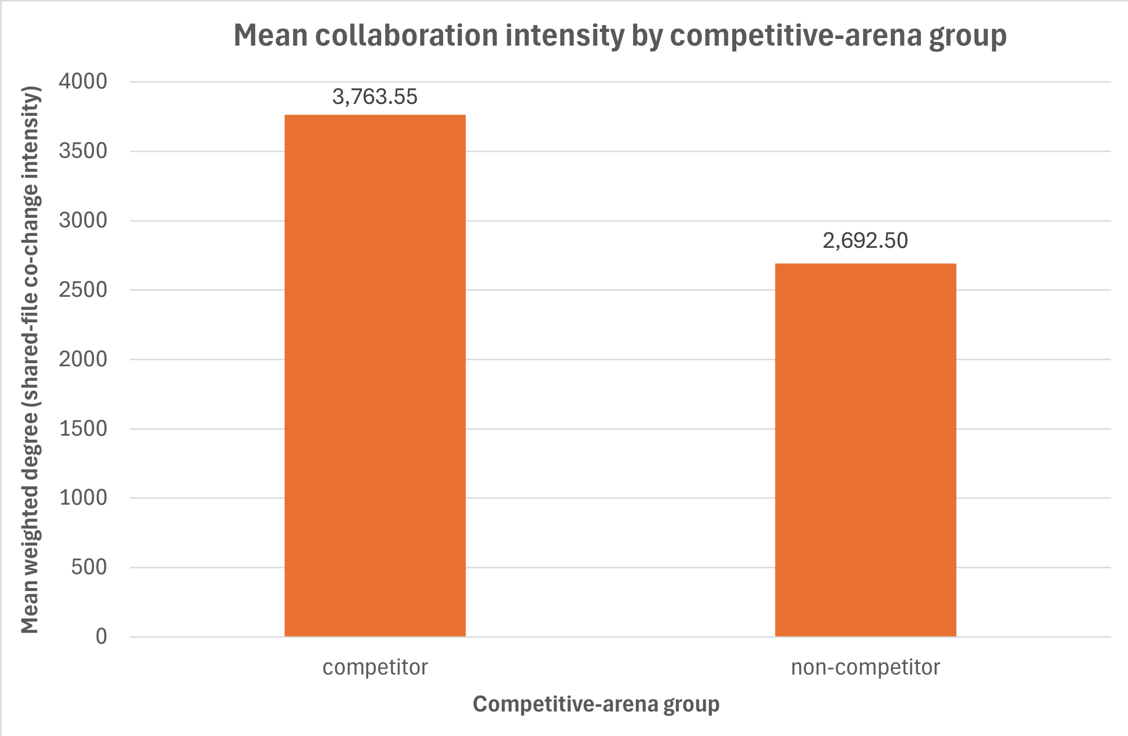
Because competitive-arena classification is available only for the manually annotated focal subset, all analyses in this section are restricted to those 54 firms. Structural metrics are still computed on the full firm-level network, but competitor-status comparisons are made only within the annotated subset.

4.3.1 Firm-Level Differences by Competitive-Arena Status

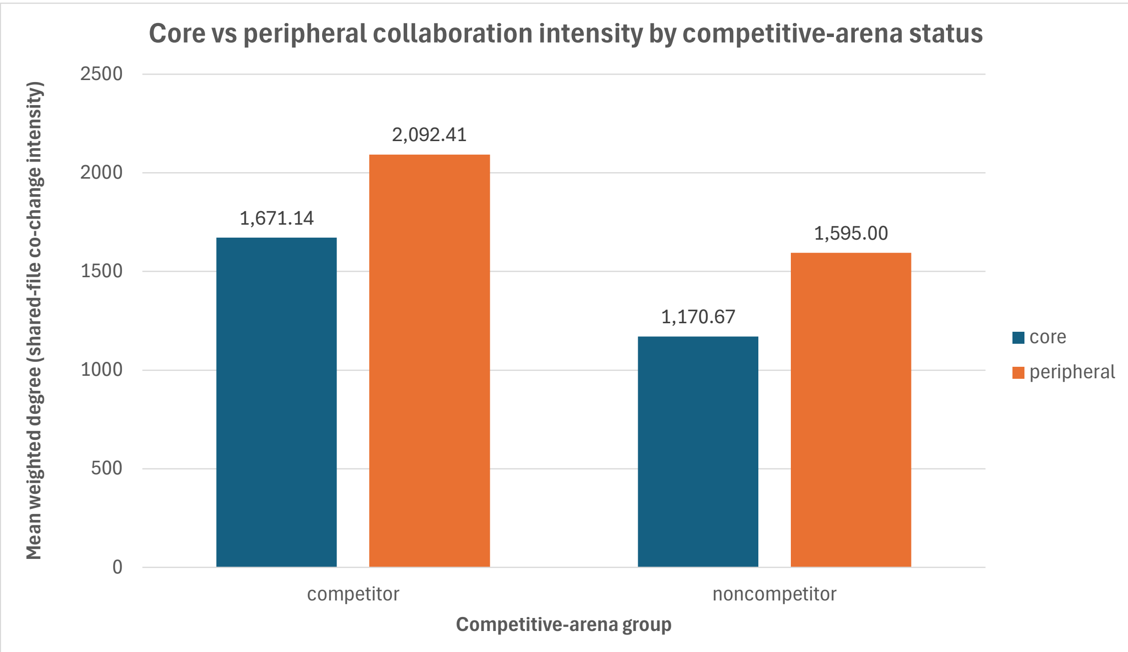
The analysis first compares collaboration intensity at the firm level. Figure 4-14 summarizes the firm-level comparison by competitive-arena status. Panel (a) reports mean weighted degree for competitor and non-competitor firms within the annotated subset in the overall network, while panel (b) shows the same comparison separately for the core and peripheral network variants.

As in Figure 4-14 Panel (a) shows, competitor firms exhibit higher mean collaboration intensity than non-competitors in the overall network (3,763.55 vs. 2,692.50). Because weighted degree is computed on the full firm-level network before filtering, these values reflect each firm's overall level of collaboration. Among the focal firms, those classified as direct competitors are, on average, more heavily involved in shared-file collaboration.

Panel (b) shows that this pattern also holds across the two scope variants. Competitors remain more collaboration-intensive than non-competitors in both the core and peripheral networks. Although absolute values are slightly higher in the peripheral network for both groups, the ordering remains unchanged. Taken together, the two panels point to the same firm-level result: within the annotated subset, competitor status is associated with higher collaboration intensity not only in the overall network, but also across both scope variants.



(a) Overall network



(b) Core vs. peripheral network variants

Figure 4-14: Firm-level collaboration intensity by competitive-arena status in the annotated subset: (a) overall network; (b) core vs. peripheral network variants.

4.3.2 Edge-Level Collaboration by Competitive Pairing

Since coopetition involves relationships between firms, the analysis next shifts from firm-level averages to collaboration between pairs of firms. Edges are categorized according to the competitive status of both firms. This analysis is restricted to edges where both endpoints belong to the annotated subset, ensuring that competitor labels are used only where they were explicitly coded. Figure 4-15 reports mean edge weight by pairing type for annotated firm pairs in the overall network.

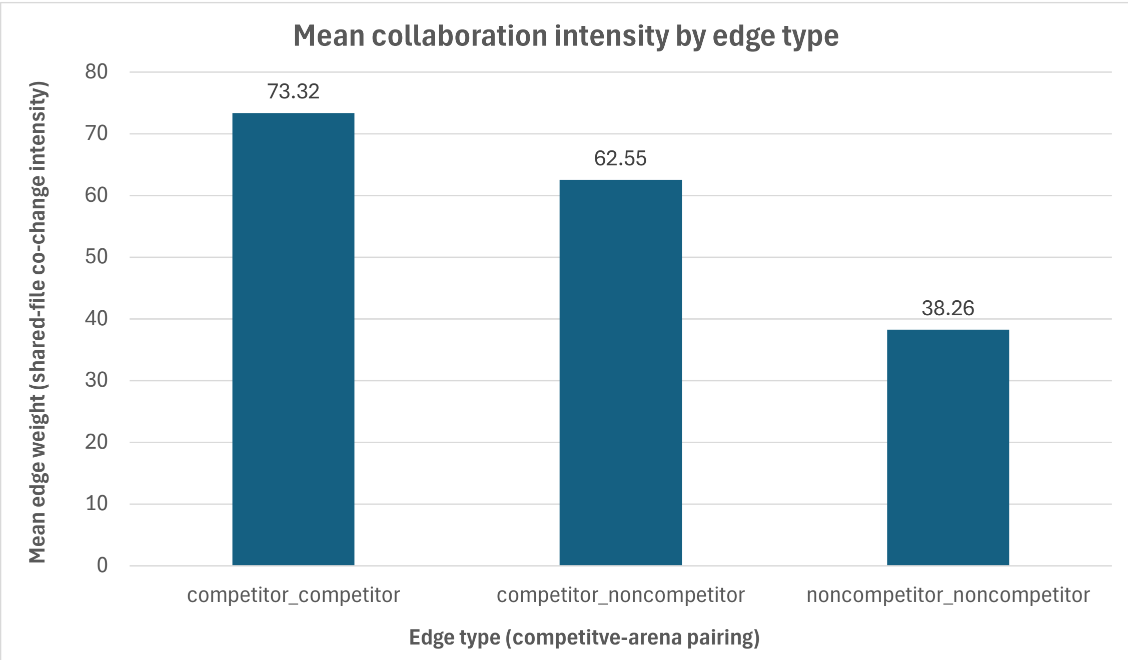


Figure 4-15: Mean collaboration intensity by edge type (annotated subset).

The competitor–competitor edges exhibit the highest mean collaboration intensity (73.32), followed by competitor–noncompetitor ties (62.55), while noncompetitor–noncompetitor ties display the lowest mean intensity (38.26). This ordering suggests that the strongest pairwise collaboration in the annotated subset is concentrated among firms that overlap competitively.

Because mean values can be influenced by extreme observations, Figure 4-16 presents median edge weights for the same edge categories.

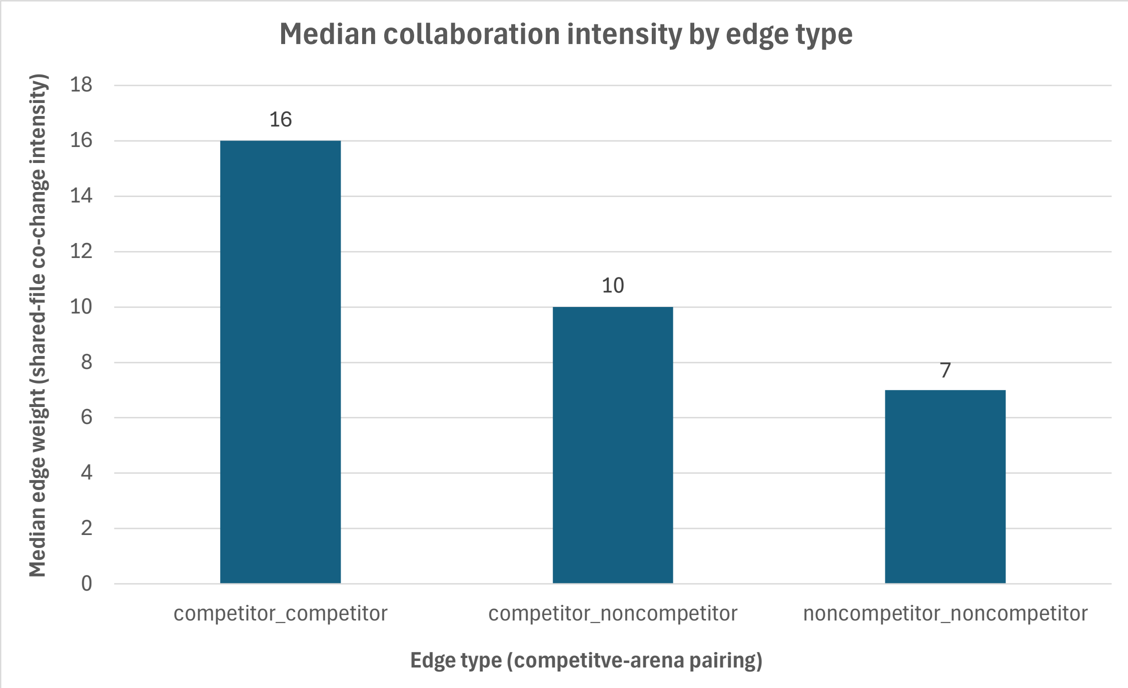


Figure 4-16: Median collaboration intensity by edge type (annotated subset).

The same ordering persists at the median level (16 for competitor–competitor, 10 for competitor–noncompetitor, and 7 for noncompetitor–noncompetitor), indicating that the pattern is not driven solely by a small number of exceptionally strong ties. Taken together, Figure 4-15 and Figure 4-16 show that competitor–competitor ties are consistently more intensive than mixed ties and noncompetitor–noncompetitor ties.

4.3.3 Scope Comparison of Competitive Pairings

To assess whether these patterns vary across repository layers, Figure 4-17 (p. 71) compares mean edge intensity separately for the core and peripheral networks, again restricted to annotated firms. As Figure 4-17 shows, competitor–competitor ties remain the most intensive category in both scope variants, followed by mixed pairings, with noncompetitor–noncompetitor ties consistently weakest. Peripheral collaboration exhibits slightly higher mean intensity values than core collaboration in all three edge categories, but the relative ordering of edge types remains stable. This indicates that the stronger collaboration observed among competitors is not confined to one specific part of the repository. Instead, the same comparative pattern appears both in the more narrowly defined core scope and

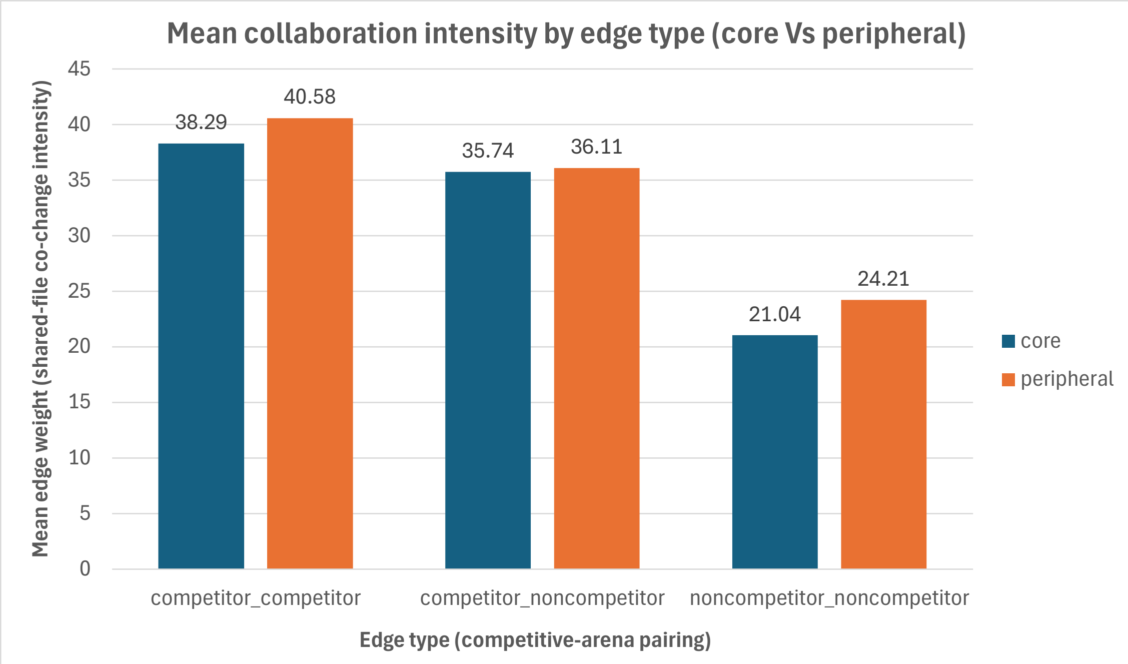


Figure 4-17: Mean collaboration intensity by edge type (core vs. peripheral, annotated subset).

in the broader peripheral layer. The scope split therefore changes the absolute level of collaboration intensity, but not the relative ordering of competitive pairings.

To check whether the same pattern also holds in the middle of the distribution, Figure 4-18 (p. 72) reports the corresponding median values. This comparison is useful because mean values can still be influenced by a smaller number of stronger ties within each category. If the same ordering persists at the median level, the pattern is more likely to reflect a broader structural tendency rather than being driven only by a few high-intensity observations.

Figure 4-18 points to the same substantive takeaway. The ordering of competitor–competitor, competitor–noncompetitor, and noncompetitor–noncompetitor ties remains unchanged at the median level across both scope variants. Peripheral median values are again slightly higher than core values, but the basic coopetitive ordering is stable across both layers. Taken together, the mean and median comparisons show that scope affects the absolute magnitude of tie intensity more than the relative structure of competitive pairings. In other words, the core and peripheral partitions differ in level, but they tell the same substantive story about which types of firm pairs collaborate most intensively.

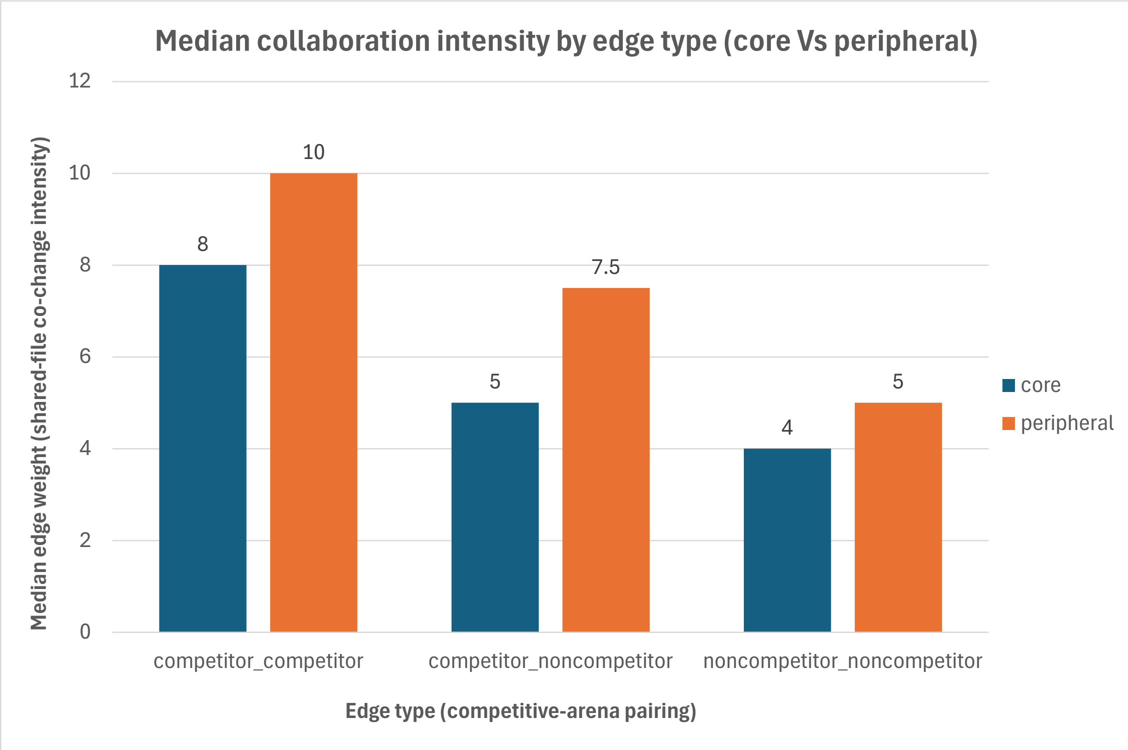


Figure 4-18: Median collaboration intensity by edge type (core vs. peripheral, annotated subset).

4.3.4 Summary of RQ3

Within the annotated focal subset, direct competitors are not less collaborative than non-competitors. At the firm level, competitors exhibit higher average collaboration intensity in the overall network, and this difference remains visible across both core and peripheral scopes. At the level of firm pairs, competitor–competitor ties exhibit the highest mean and median collaboration intensity, followed by mixed pairings, whereas noncompetitor–noncompetitor ties are comparatively weaker. The consistency of this ordering across scope variants suggests that the pattern is not limited to a single repository layer.

These findings indicate that competitive proximity does not suppress upstream collaboration among the most actively engaged firms in the ecosystem. Rival firms appear to share substantial technical overlap in development activity. One plausible interpretation is that firms competing in related downstream arenas are also more likely to depend on overlapping upstream framework components and therefore to converge on a common set of repository artifacts. This helps explain why competitor–competitor ties are stronger than mixed ties and noncompetitor–noncompetitor ties: firms with greater market overlap may also face more similar technical requirements and therefore engage more intensely with the same parts of the shared framework. In that sense, competitive overlap may coexist with, and may even be associated with, stronger technical co-presence around the shared technological base.

This does not imply the absence of competition. Rather, it points to a structural separation between shared infrastructure development and downstream product differentiation. Firms that compete in adjacent market arenas nonetheless co-modify the same technical artifacts, contributing to and depending on a common technological base. The empirical results therefore support interpreting PyTorch as a coopetitive infrastructure ecosystem in which collaboration around shared foundations persists despite competitive overlap in commercial domains.

4.4 Chapter Synthesis

Repository mining combined with social network analysis provides an empirical representation of inter-firm collaboration in the PyTorch ecosystem. The reconstructed firm-level network is not evenly distributed, but marked by concentration, hierarchical differentiation, and scope-sensitive variation across the core and peripheral network variants. A relatively small number of firms account for a disproportionate share of shared artifact engagement, while many others remain at substantially lower levels of collaboration intensity. In this sense, the analysis shows that repository trace data can be used to reconstruct a structural view of upstream inter-firm collaboration.

Firms also do not occupy equivalent positions within that structure. The platform sponsor stands out as the dominant actor in terms of collaboration intensity and embeddedness, while infrastructure-oriented complementors and selected competitors form a second tier of strongly engaged firms. Other actors, especially more downstream and enterprise-oriented firms, participate in more selective ways. Brokerage is also unevenly distributed, with a smaller set of firms occupying more intermediary positions than others. The collaboration network therefore reflects not only uneven participation, but also differentiated forms of ecosystem involvement linked to the annotated role structure used in the study.

The competitor-based analyses further show that competitive proximity does not suppress collaboration within the focal ecosystem. Competitor firms exhibit higher mean collaboration intensity than non-competitors at the firm level, and competitor–competitor ties are stronger on both mean and median measures than mixed or noncompetitor pairings.

This ordering remains visible across the core and peripheral network variants. Firms with overlapping downstream market positions therefore also appear to converge strongly in upstream technical work around the shared framework.

Overall, the results suggest that the PyTorch ecosystem operates as a layered coopetitive infrastructure. The network is concentrated rather than flat, role-differentiated rather than uniform, and shaped by competitive overlap rather than insulated from it. The sponsor retains a structurally dominant position, but the ecosystem also depends on a broader set of complementors and competitors whose contributions vary in intensity, embeddedness, and brokerage. Stronger collaboration among competitors indicates that shared infrastructure development can remain a common area of interaction even where firms pursue differentiated downstream strategies.

The overall pattern is one of simultaneous technological interdependence and strategic differentiation. Cooperation does not remove competitive boundaries, and competition does not eliminate collaboration. Instead, both are organized through a layered structure linking sponsor dominance, differentiated ecosystem roles, and sustained upstream overlap among firms with adjacent commercial interests.

5 Discussion and Contributions

5.1 Discussion

This chapter interprets the findings reported in Chapter 4 in light of the theoretical framework developed in Chapter 2. Within the boundaries of this study, the PyTorch case indicates that inter-firm coopetition in vendor-led open-source ecosystems is better understood as a structurally differentiated pattern of simultaneous cooperation and competition than as a uniform or purely dyadic relationship. The reconstructed network shows that firms do not participate in the ecosystem to the same extent. Instead, collaboration is concentrated, role-dependent, and shaped by the asymmetries of a sponsor-led governance setting. This is in line with the broader coopetition perspective outlined in Chapter 2, according to which firms may cooperate and compete at the same time, but do so from unequal positions within a broader ecosystem structure [BK00, BK14, CSL+20].

One implication of the findings concerns sponsor dominance in vendor-led OSS ecosystems. Chapter 2 described such ecosystems as settings in which one host or sponsor often retains disproportionate influence over development direction, governance, and the structuring of participation. The PyTorch case fits that pattern. The platform sponsor is not simply one participant among many, but the structurally dominant actor in the collaboration network. This suggests that vendor-led coopetition should not be viewed only through a symmetric rivalry lens. Instead, the findings point to a form of coopetition in which the host firm occupies a privileged structural position while other firms participate within conditions set by that asymmetry. At the same time, cooperation around the shared code base does not imply equal influence over the ecosystem. The host may remain the main structural hub while external firms still contribute intensively to the shared technological base. Upstream collaboration can therefore coexist with concentrated structural power, and formal openness of participation does not remove hierarchy [ODH+24].

The findings also support the layered character of coopetition. Chapter 2 emphasized that open-coopetition in OSS ecosystems often involves a structural separation between shared upstream development and downstream commercial differentiation. The findings from Chapter 4 fit that logic. Firms classified as direct competitors of the sponsor are not less collaborative than non-competitors. Rather, the observed pattern suggests that competitive overlap in downstream markets can coexist with strong upstream technical co-presence around the same shared framework. This does not mean that competition disappears. It instead points to a layered pattern in which firms remain rivals in commercial arenas while still converging on the same shared infrastructure because that infrastructure is strategically important for multiple participating firms. In the theoretical terms developed in Chapter 2, this pattern is compatible with the idea that the shared framework functions as a common technological base around which multiple firms organize both complementary and rival strategies. The PyTorch case therefore supports viewing vendor-led OSS ecosystems as coopetitive infrastructure arenas. In such arenas, collaboration persists because firms depend on overlapping upstream components, while competition persists because those same firms pursue differentiated downstream value capture [CSL+20, TAL+25]. This reading is broadly compatible with the commoditization logic discussed in Chapter 2, although the present design does not test such strategies directly [VLM09].

The findings also highlight ecosystem role differentiation. Chapter 2 argued that OSS and platform ecosystems are composed of heterogeneous actors such as sponsors, complementors, rivals, and service-oriented participants, and that their positions in the ecosystem are strategically meaningful. The empirical analysis supports that view. The network is not only concentrated around the sponsor, but also clearly differentiated among non-sponsor firms. Infrastructure-oriented actors, especially those closer to hardware, compute, cloud, or integration functions, occupy stronger structural positions than more downstream or application-oriented participants. This suggests that the annotated ecosystem roles capture meaningful differences in structural positioning within the collaboration network [ODH+24, TAL+25]. More broadly, participation should not be treated as a binary matter of whether a firm contributes or not. Firms instead participate through differentiated structural roles that vary in intensity, embeddedness, and brokerage. This supports the view that coopetition in ecosystem settings is a multi-actor and network-level phenomenon rather than only a dyadic one, and that vendor-led open-source ecosystems are better understood as layered strategic networks than as flat collaboration spaces [BN95, BK14, CSL+20].

The findings also suggest that network position is a strategically relevant feature of ecosystem participation. Chapter 2 connected coopetition theory to social network arguments about embeddedness, brokerage, and network-enabled advantage. The present findings support the usefulness of that connection at an interpretive level. Firms with strong positions in the PyTorch network are not randomly distributed across the ecosystem. They are concentrated among the sponsor and a relatively small set of infrastructure-oriented actors. This suggests that network position provides a useful structural lens for examining how firms are situated within a shared technological base. At the same time, the study does not claim that structural position alone explains firm performance, strategic intent, or causal advantage. The point here is narrower: the case shows that repository-derived network structure can make ecosystem asymmetries empirically visible in a way that is theoretically meaningful [ZB05, Lav06].

The comparison between core and peripheral repository areas adds a further refinement. The findings suggest that inter-firm collaboration is not organized in the same way across all parts of the repository. Core collaboration is more sharply concentrated and more strongly hierarchical, whereas peripheral collaboration remains unequal but is somewhat more broadly distributed. This suggests that coopetition in vendor-led OSS ecosystems is not only actor-differentiated, but also sensitive to repository scope. Shared development around more foundational subsystems appears more concentrated among a smaller set of highly engaged firms, while surrounding layers allow broader participation. This does not establish a definitive hierarchy of strategic importance for all repository artifacts, but it does support the broader idea that different layers of shared infrastructure may attract different forms of inter-firm involvement.

Overall, the PyTorch case supports an interpretation of vendor-led OSS ecosystems in which coopetition is structured by sponsor asymmetry, differentiated ecosystem roles, and sustained upstream overlap among firms that may still compete downstream. The case does not justify universal claims about all open-source ecosystems, nor does it resolve the internal motives behind firm participation. However, within the boundaries of this design, it indicates that vendor-led open-source ecosystems can operate as layered coopetitive infrastructures in which collaboration and competition are organized across different levels and domains of the same ecosystem [ODH+24, TAL+25, CSL+20].

5.2 Validity Boundaries and Interpretation Limits

The findings of this thesis should be interpreted within the operational boundaries of the research design. Most importantly, the reconstructed network is based on shared-file co-modification and therefore captures structural co-presence around repository artifacts rather than direct social coordination. An observed tie indicates that two firms modified the same file within the observation window, while a stronger tie indicates broader shared engagement across distinct files. This provides a meaningful proxy for upstream technical overlap, but it is not direct evidence of contractual partnership, interpersonal communication, joint planning, or formal alliance formation. The discussion of coopetition in this thesis should therefore be read as a structural interpretation of repository-based collaboration patterns rather than as a direct observation of firms' internal coordination processes.

A further boundary concerns organizational attribution. Firm-level nodes are constructed only from contributors classified as *firm_affiliated*, after deterministic identity stabilization and rule-based affiliation inference. This improves consistency and transparency, but it also narrows the scope of inference. Contributors in residual or uncertain categories are excluded from firm-node construction, and affiliation is held static over the observation window once assigned at contributor level. As a result, the network describes inter-firm collaboration among contributors with sufficiently credible firm attribution rather than the full social ecology of the PyTorch project. It also means that employer changes occurring within the observation period are not modeled dynamically. The resulting firm-level graph is therefore best understood as a conservative organizational reconstruction rather than a complete census of all possible contribution relationships.

The core-peripheral distinction introduces a further interpretation limit. The study uses deterministic file-path rules to separate selected foundational subsystems from surrounding repository layers. This provides a transparent and reproducible analytical partition, but it should not be interpreted as a definitive measure of true architectural centrality or strategic importance. Some files outside the selected core paths may still be highly important, while some files inside them may involve routine maintenance. Accordingly, the distinction between core and peripheral networks is an analytical operationalization that enables structured comparison under a fixed rule, not a claim to reproduce the full architecture of PyTorch. Because the overall, core, and peripheral networks are constructed using the same tie-formation logic and the same metric definitions, differences across these variants should be interpreted as differences in artifact scope rather than as artifacts of changing network construction rules. The discussion of scope sensitivity should therefore remain tied to this explicit operational definition.

The annotation layer also places a boundary on inference. The detailed role and competitor classifications apply only to the focal subset of 54 firms. These annotations are theory-guided and are used as an interpretive overlay after network construction and metric computation. They do not alter the topology of the graph, but they do shape how subgroup differences are interpreted. At the same time, the structural measures themselves are computed on the complete network before subgroup-based interpretation, which reduces the risk that the observed differences are simply artifacts of restricting the analysis to annotated firms. Because the competitive-arena classification exists only for the annotated subset, conclusions about competitor and non-competitor patterns should be read as applying to that focal set of firms rather than to every actor in the network. In addition, the competitor variable is a simplified binary indicator of downstream market overlap with the sponsor and cannot capture all gradations or changes in competitive proximity over time.

The temporal and contextual scope of the study places an additional limit on interpretation. The analysis focuses on one repository, one branch history, and one observation window. It reconstructs an aggregate structural network over that period rather than modeling how collaboration changes over time. Although this is appropriate for the aims of the thesis, it limits the claims that can be made. The findings do not show how coopetition dynamics evolved year by year, how governance shifts affected particular phases of the ecosystem, or whether the same patterns would appear in other vendor-led OSS projects. The case therefore supports analytical generalization to theory more than statistical generalization across ecosystems. It is useful for refining how coopetition in vendor-led OSS ecosystems may be conceptualized, but it does not by itself establish that all such ecosystems will display the same structure.

Relatedly, the present design supports structural interpretation rather than causal explanation. The network metrics describe collaboration intensity, embeddedness, and brokerage within the reconstructed graph, but they do not identify why firms occupy these positions or what organizational outcomes those positions produce. For example, stronger collaboration among competitors can be interpreted as evidence of overlapping upstream dependence on the shared framework, but the design cannot show that this is the only mechanism behind the pattern. Nor can it determine the extent to which firm strategy, internal governance, resource asymmetries, or individual developer practices produced the observed structure. The discussion therefore remains intentionally cautious: the findings suggest theoretically compatible interpretations, but they do not establish causal mechanisms directly.

These boundaries do not undermine the value of the study. Rather, they define the level at which its conclusions are valid. The thesis provides a transparent and reproducible structural account of inter-firm collaboration in a vendor-led OSS ecosystem and interprets that structure through a coopetition lens. Its claims should therefore remain within that level of analysis: repository-based, firm-level, structurally interpreted, and theoretically informed, but not extended beyond what the design can support.

5.3 Contributions

This thesis makes contributions at the theoretical, methodological, and empirical levels.

At the theoretical level, it contributes to the study of coopetition in vendor-led open-source ecosystems. As discussed in Chapter 2, the existing literature emphasizes the simultaneous presence of cooperation and competition, sponsor asymmetry, role differentiation, and the importance of shared technological infrastructures in such settings [BK00, BK14, CSL+20, ODH+24, TAL+25]. This thesis brings these elements together through a network-level interpretation of vendor-led OSS coopetition. Within the boundaries of the PyTorch case, the findings support a view of vendor-led OSS ecosystems as layered coopetitive infrastructures in which a sponsor retains structural dominance, non-sponsor firms occupy differentiated roles, and direct competitors may still collaborate intensely around a shared upstream technological base. In this way, the thesis refines the discussion of coopetition by moving beyond a flat or purely dyadic understanding and by showing how cooperation and competition can be structurally organized within the same ecosystem.

At the methodological level, the thesis develops a transparent repository-mining and social network analysis workflow for studying inter-firm coopetition from trace data. The design links commit-file traces, deterministic identity stabilization, conservative affiliation inference, firm-level aggregation, scope-based network partitioning, and theory-guided annotation into one workflow. In doing so, it addresses part of the gap identified in Chapter 2 between high-level coopetition theory and scalable empirical analysis in OSS settings. The methodological contribution is not that repository traces capture every aspect of strategy, but that they can be used to reconstruct a reproducible structural map of inter-firm technical co-presence that supports theory-informed interpretation. The explicit separation between computationally derived network structure and later interpretive annotation is also important because it preserves transparency about what is directly measured and what is added conceptually.

A further methodological contribution lies in the introduction of the core-versus-peripheral scope comparison under constant tie-construction logic. Rather than treating the repository as a single flat space of collaboration, the study shows how a deterministic path-based partition can be used to compare structural patterns across different repository layers without changing the underlying network model. This does not solve the broader problem of measuring strategic contribution types perfectly, but it offers a practical and reproducible way to add analytical differentiation to repository-based inter-firm network analysis. In that sense, the thesis offers a middle ground between flat network reconstruction and purely qualitative interpretation.

At the empirical level, the thesis provides a detailed firm-level analysis of the PyTorch ecosystem as a vendor-led open-source AI infrastructure project. This is valuable because PyTorch is a strategically important case and because the study examines it directly rather than using it only as a side reference within broader discussions of AI ecosystems. Empirically, the thesis shows that the inter-firm collaboration structure in PyTorch is concentrated, hierarchical, scope-sensitive, and role-differentiated. It also shows that direct competitors of the sponsor are not structurally peripheral by default and may exhibit substantial collaboration intensity around the shared framework. These findings add empirical evidence to the literature on company-hosted and vendor-led OSS ecosystems, particularly in AI-related infrastructure contexts [ODH+24, TAL+25].

At the same time, these contribution claims should remain proportionate to the design. This thesis does not claim to provide a general theory of all OSS ecosystems, nor does it claim to identify firms' internal strategic motives with certainty. Its contribution is narrower. It offers a theory-informed, empirically grounded, and methodologically transparent account of how inter-firm coopetition can be reconstructed and interpreted in one important vendor-led OSS ecosystem. In doing so, it helps clarify how sponsor asymmetry, role differentiation, and upstream technical overlap among rivals can be made visible through repository-based network analysis.

Overall, the thesis contributes to the study of inter-firm coopetition in vendor-led OSS ecosystems by connecting strategy theory, repository mining, and social network analysis in one design. It shows that large-scale repository traces can be used not only to describe development activity, but also to make visible structural patterns of ecosystem participation, provided that the interpretation remains disciplined and consistent with the design boundaries of the study.

6 Conclusion

This thesis examined how inter-firm coopetition can be studied in a vendor-led open-source ecosystem through repository mining and social network analysis. Using PyTorch as the empirical case, it addressed three related questions: how repository-based network reconstruction can model the collaborative dimension of inter-firm coopetition, how firms' structural positions differ within that network, and how collaboration patterns vary between direct competitors and non-competitors.

The findings show that repository-based network analysis can reconstruct a structural view of inter-firm collaboration in PyTorch. The reconstructed network is not flat or evenly distributed, but concentrated and hierarchical, with clear differences between the overall, core, and peripheral scope variants. A relatively small number of firms account for a disproportionate share of shared artifact engagement, and collaboration around more foundational subsystems appears more concentrated than collaboration in surrounding repository layers. This suggests that inter-firm collaboration in the ecosystem is scope-sensitive rather than uniform.

The analysis also shows that firms do not occupy similar positions within this structure. The platform sponsor is the dominant structural actor, while non-sponsor firms differ in collaboration intensity, embeddedness, and brokerage. Infrastructure-oriented firms are positioned more strongly than many downstream-oriented actors, which suggests that ecosystem participation is differentiated and layered rather than homogeneous. This supports the interpretation that inter-firm coopetition in vendor-led OSS ecosystems unfolds through unequal structural positions rather than symmetric participation.

The findings also clarify the relationship between collaboration and competition. Firms classified as direct competitors of the sponsor are not less collaborative than non-competitors. Instead, stronger collaboration among competitors suggests that downstream rivalry does not eliminate upstream interdependence around the shared framework. Within the boundaries of this design, the PyTorch case points to a layered form of coopetition in which firms may continue to compete in surrounding markets while remaining connected through the development of shared infrastructure.

Taken together, the findings indicate that vendor-led OSS ecosystems can be understood as layered coopetitive infrastructures marked by sponsor asymmetry, differentiated ecosystem roles, and sustained upstream overlap among firms with adjacent commercial interests. At the same time, the thesis remains cautious about the level of inference that this design can support. The reconstructed network captures structural co-presence around shared repository artifacts rather than direct coordination, the firm-level graph depends on attribution rules and scope definitions, and the case design does not establish causal mechanisms or universal generalizations across all OSS ecosystems.

On this basis, the thesis contributes at three levels. At the theoretical level, it refines the understanding of coopetition in vendor-led OSS ecosystems by showing how sponsor dominance, role differentiation, and rival collaboration around shared infrastructure can be interpreted together at the network level. At the methodological level, it develops a transparent pipeline that links repository traces, firm attribution, structural network reconstruction, scope comparison, and theory-guided annotation. At the empirical level, it provides a detailed firm-level analysis of the PyTorch ecosystem as a strategically important AI infrastructure project.

Future research could extend this work in three directions. First, comparative analysis across multiple vendor-led OSS ecosystems could show whether similar structural patterns appear in other settings. Second, temporal analysis could examine how coopetition changes across governance phases, release cycles, or strategic turning points. Third, repository-based structural analysis could be combined with qualitative or organizational data to better connect observed network positions with firm motives, governance processes, and downstream outcomes.

Overall, this thesis shows that inter-firm coopetition in a vendor-led OSS ecosystem can be studied through repository mining and social network analysis. By focusing on PyTorch, it shows that collaboration among firms in open-source ecosystems is neither flat nor neutral, but structured by hierarchy, role differentiation, and competitive overlap. Within the limits of the design, the study provides an empirical basis for understanding how coopetition operates in strategically important shared software infrastructures.

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Documentation of AI Tool Use

During the preparation of this thesis, ChatGPT and Grammarly were used in a supportive capacity for paraphrasing, wording refinement, readability improvement, and language revision. DeepL was used only for translation support in relation to the German abstract.

The use of these tools was limited to linguistic and editorial support. They were not used as scientific sources. All substantive arguments, interpretations, methodological decisions, and the final wording of the thesis remain solely the responsibility of the author.

A Annotated Firms and Role Labels

This appendix lists the focal firms included in the detailed manual annotation subset used in the empirical analysis. For each firm, the table reports the assigned fine-grained `role_label`, `firm_type`, and `direct_competitor` classifications.

Table 1-1: Annotated firms and assigned fine-grained role labels (N = 54).

Firm	**role_label**	**firm_type**	**direct_competitor**
Meta	ecosystem_orchestrator	platform_sponsor	0
Nvidia	core_hardware_complementor	gpu_vendor	1
Cfcs	broker_integrator	integrator_consultant	0
Openai	foundation_model_lab	ai_lab	1
Quansight	broker_integrator	integrator_consultant	0
Intel	core_hardware_complementor	gpu_vendor	1
Amd	core_hardware_complementor	gpu_vendor	1
Microsoft	cloud_platform_complementor	cloud_provider	1
Ibm	cloud_platform_complementor	cloud_provider	1
Amazon	cloud_platform_complementor	cloud_provider	1
Leptonai	infra_ai_platform	cloud_provider	1
Anthropic	foundation_model_lab	ai_lab	1
Apple	bigtech_enterprise_user	enterprise_user	0
Huawei	telco_hw_enterprise_user	enterprise_user	0
Google	cloud_platform_complementor	cloud_provider	1
Rune	peripheral_firm_unknown	other	0
Preferred Networks	specialised_ai_lab	ai_lab	1
Graphcore	core_hardware_complementor	gpu_vendor	1
Fujitsu	diversified_tech_user	enterprise_user	0
Cruise	automotive_ai_user	enterprise_user	0
Two Sigma Investments	financial_ai_user	enterprise_user	0
Kitware	technical_services_firm	integrator_consultant	0
Astral-Sh	peripheral_firm_unknown	other	0
Bytedance	consumer_ai_platform_user	enterprise_user	0
Offscale	peripheral_firm_unknown	other	0

Firm	**role_label**	**firm_type**	**direct_competitor**
Lightning Ai	mlops_platform_integrator	integrator_consultant	0
Deshaw	financial_ai_user	enterprise_user	0
Protopia	ml_consulting_firm	integrator_consultant	0
Cerebras Systems	core_hardware_complementor	gpu_vendor	1
Arm	core_hardware_complementor	gpu_vendor	1
Safetykit	peripheral_firm_unknown	other	0
Near	peripheral_firm_unknown	other	0
Sharechat	consumer_ai_platform_user	enterprise_user	0
Voltrondata	data_infra_firm	integrator_consultant	0
Naver	cloud_platform_complementor	cloud_provider	1
Mts Ai	telco_ai_user	enterprise_user	0
Paige Ai	healthcare_ai_user	enterprise_user	0
Alibaba	cloud_platform_complementor	cloud_provider	1
Linkedin	bigtech_enterprise_user	enterprise_user	0
Provisio Gmbh	technical_services_firm	integrator_consultant	0
Yandex	cloud_platform_complementor	cloud_provider	1
Sakanaai	foundation_model_lab	ai_lab	1
Databricks	data_platform_provider	integrator_consultant	0
Red Hat	enterprise_software_vendor	integrator_consultant	0
Xai	foundation_model_lab	ai_lab	1
Epam Systems	technical_services_firm	integrator_consultant	0
Baidu	cloud_platform_complementor	cloud_provider	1
Perplexity Ai	ai_product_lab	ai_lab	1
Mail.Ru	cloud_platform_complementor	cloud_provider	1
Uber	mobility_ai_user	enterprise_user	0
Zoox	automotive_ai_user	enterprise_user	0
Cloudera	data_platform_provider	integrator_consultant	0
Hudson River Trading	financial_ai_user	enterprise_user	0